Roma Tre

Information Visualization

Our Visual Capabilities

M. Patrignani

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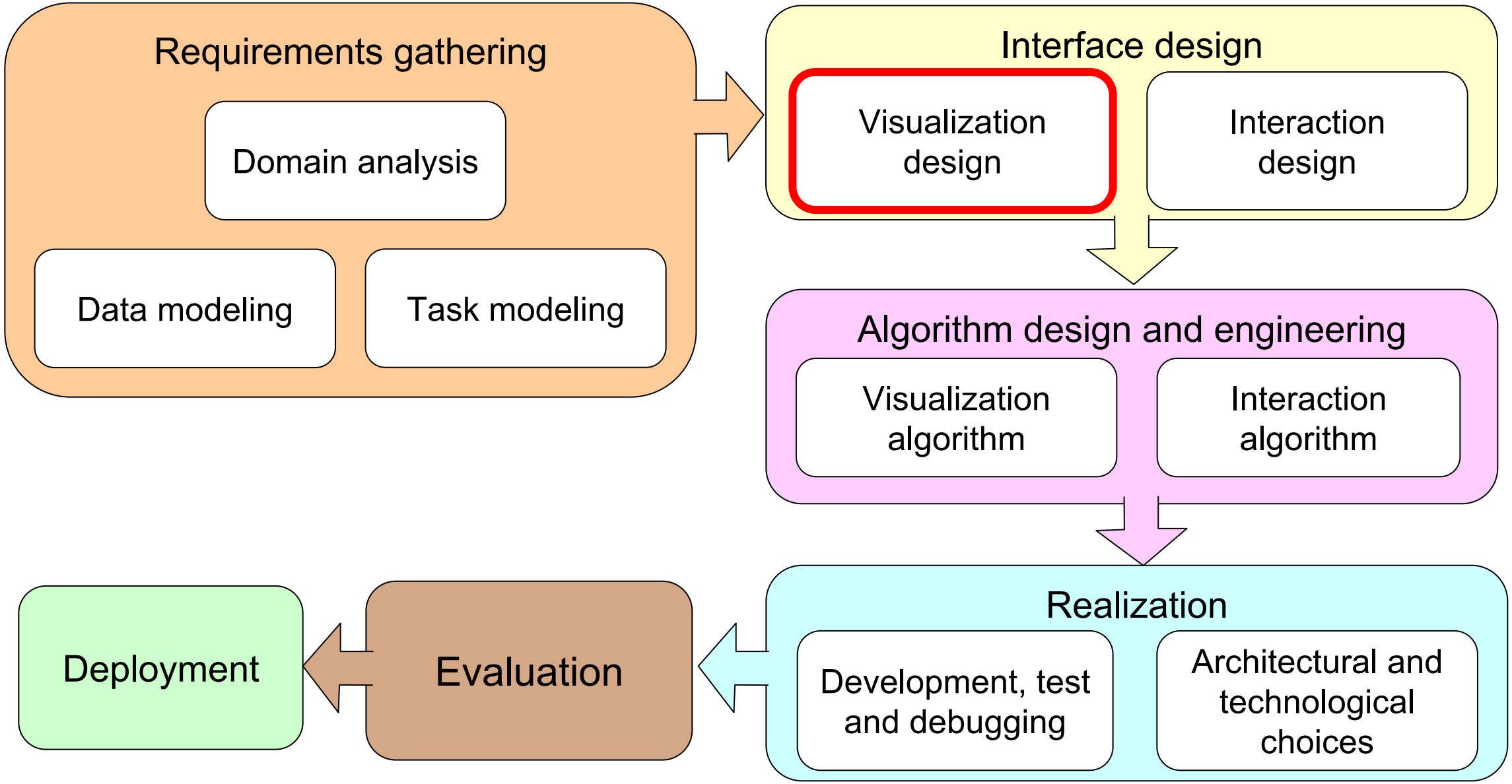
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Summary

- Our vision's principles and limitations
  - peripheral and central view
  - edge detection mechanisms
  - perception illusions explained

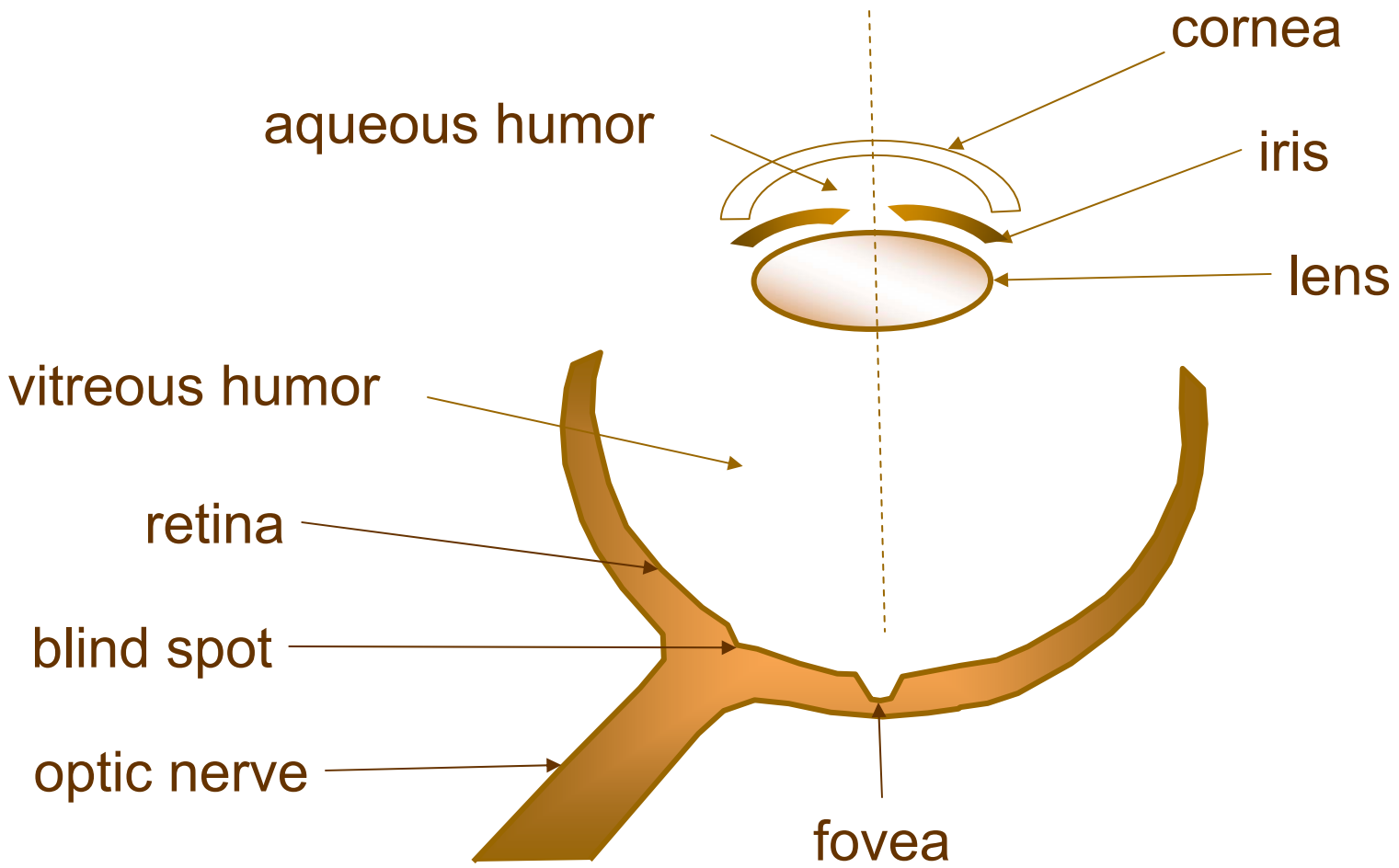
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Design pipeline



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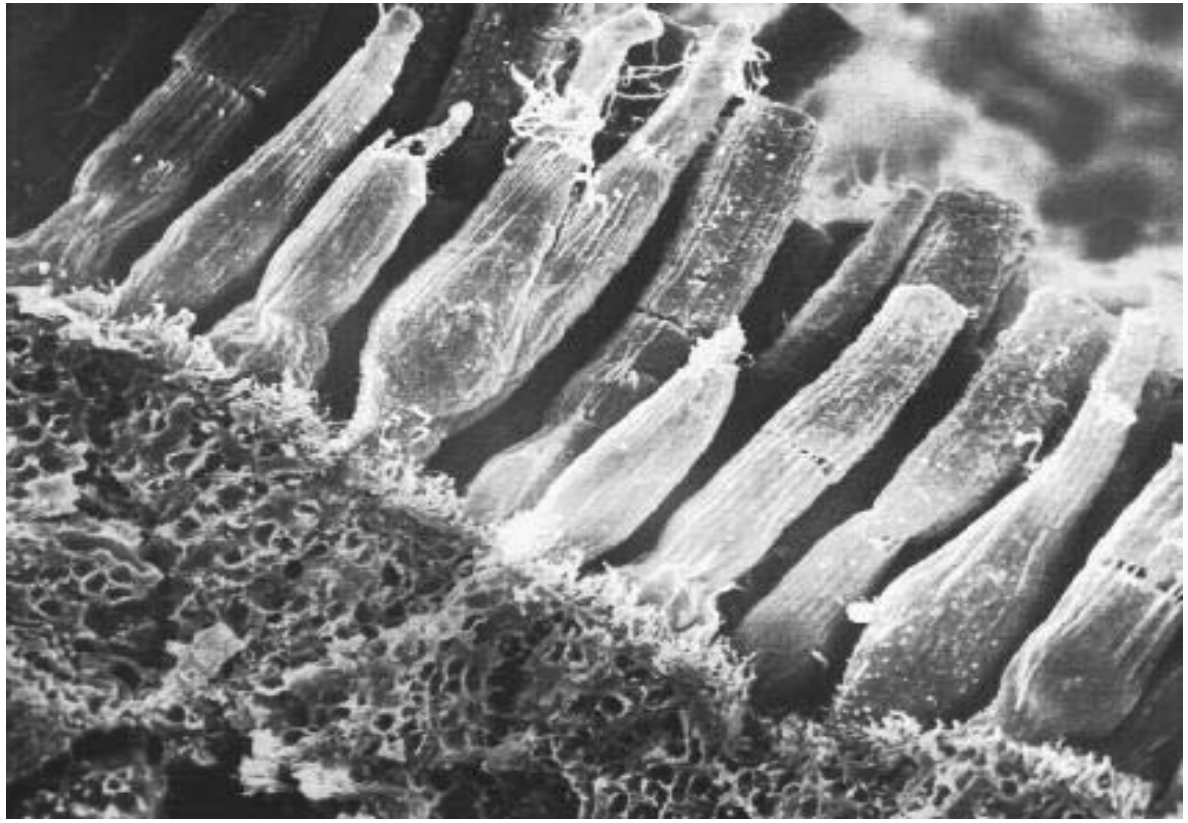
Your (right) eye

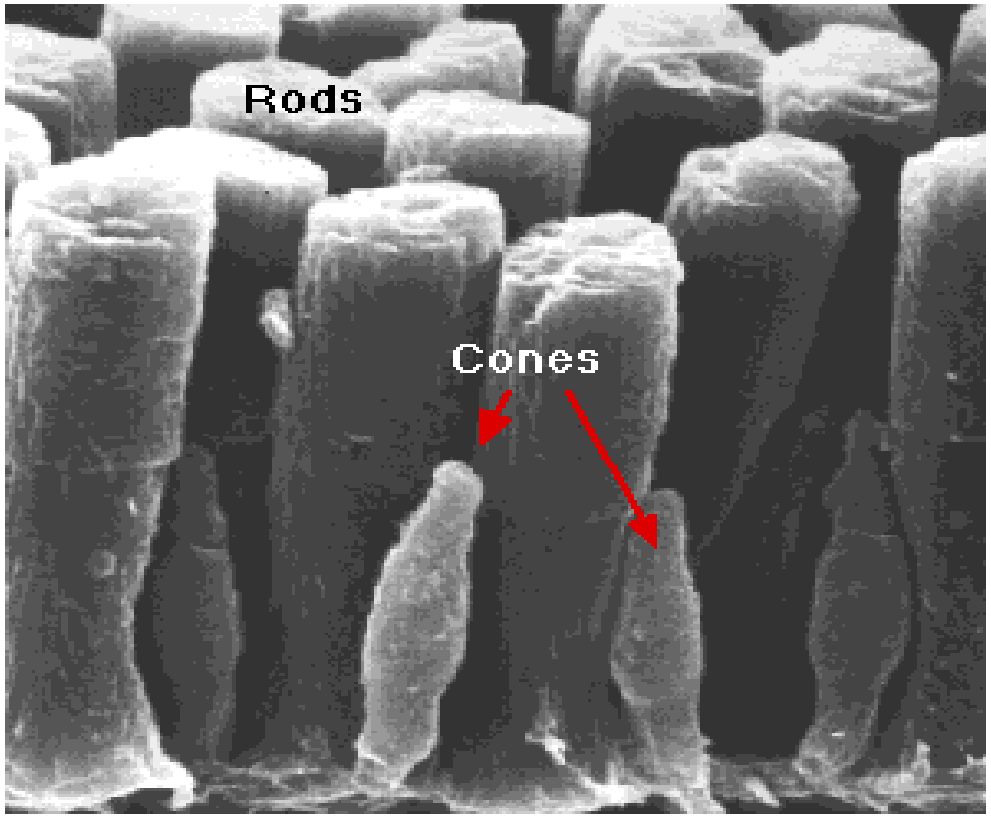


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Retina receptors

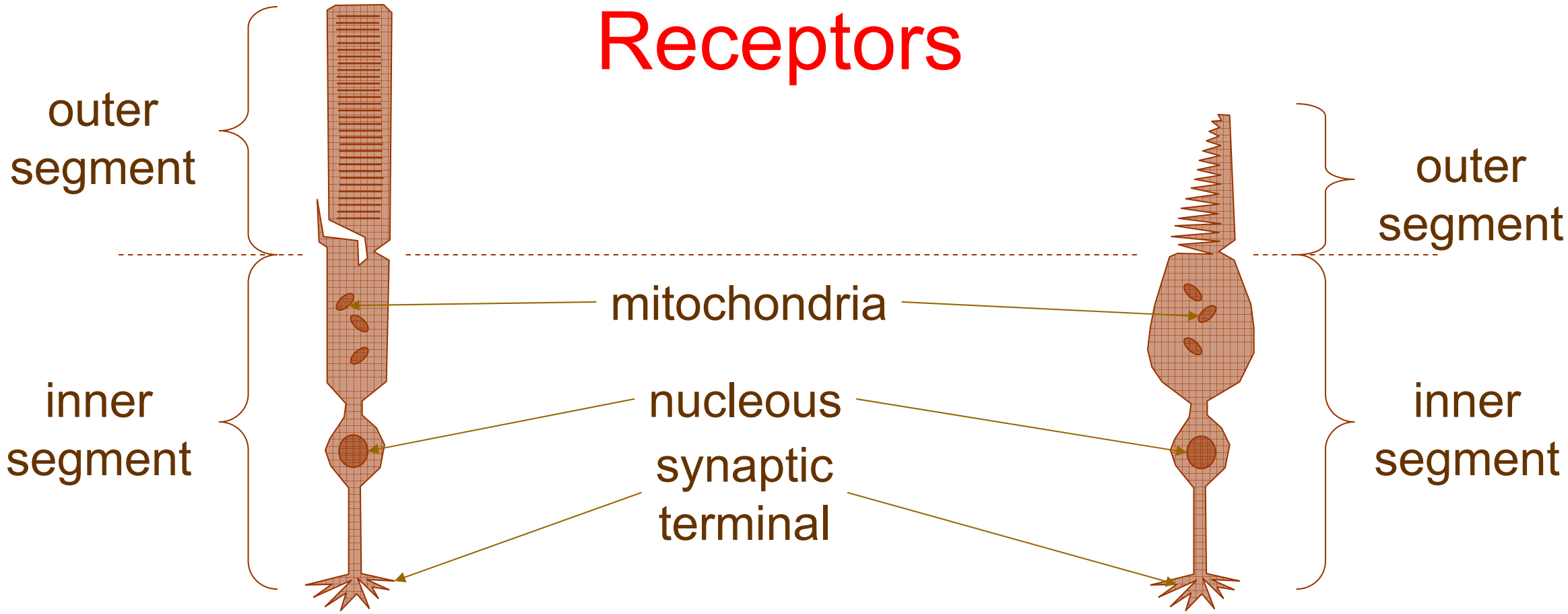
- 70% of body's sense receptors reside in our eyes





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Receptors



Rods

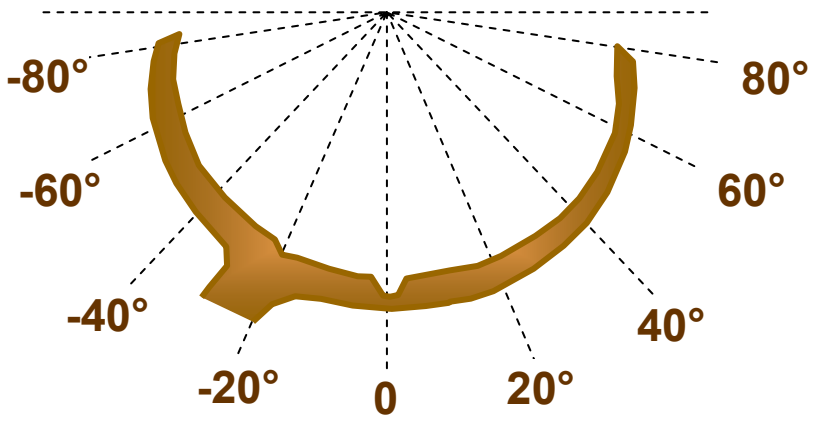
- 120 million
- responds to a single photon of light
- no color sensitivity

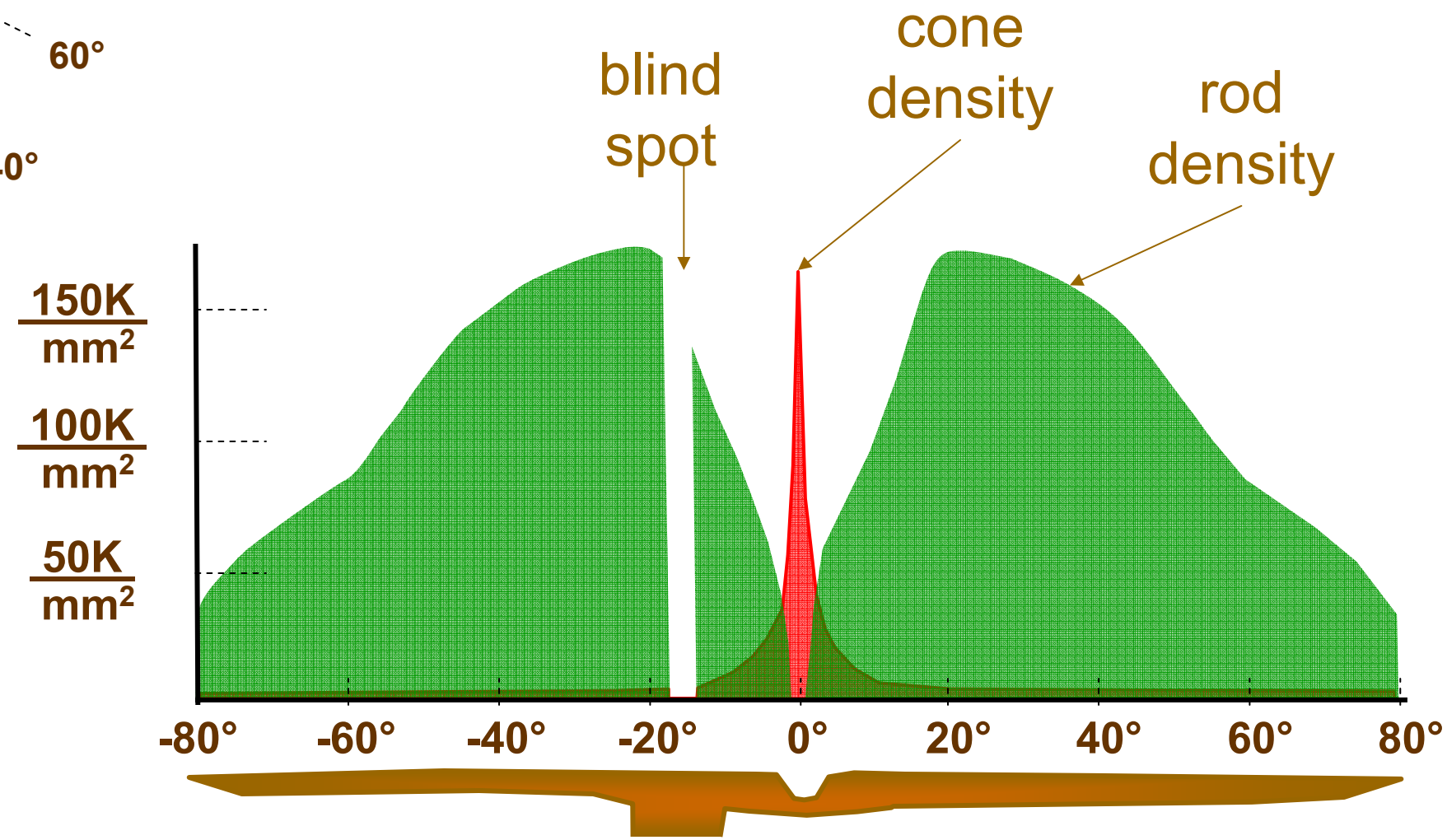
Cones

- 6-7 million
- 100 times less sensitive to light than rods
- color sensitivity

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Cone and rod densities

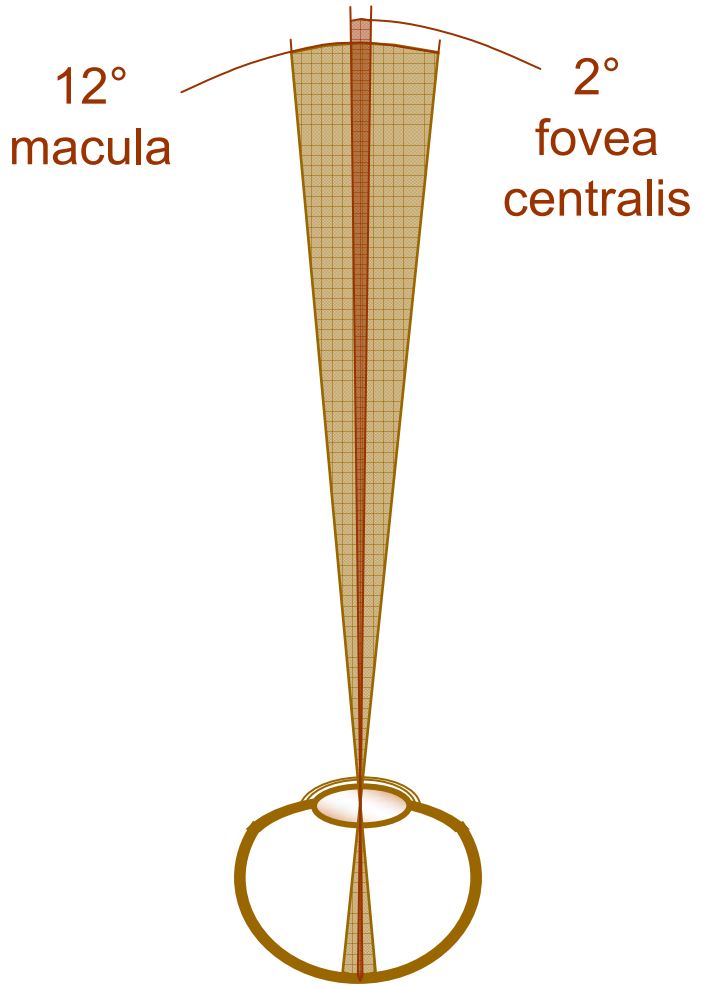




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Cone concentration

- Macula
  - central 12 degrees of visual field
  - most of the cones concentrate here
- Fovea centralis
  - central 2 degrees of visual field
    - twice the width of your thumbnail at arm's length
  - comprises less than 1% of retinal size but 50% of visual cortex



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Your eyes are not cameras

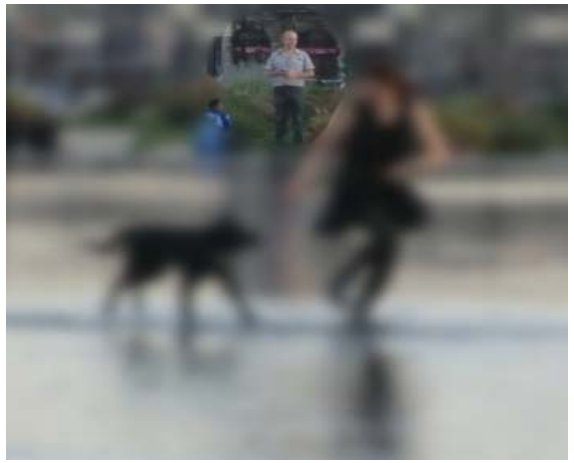


What you believe to see…

What you actually see…



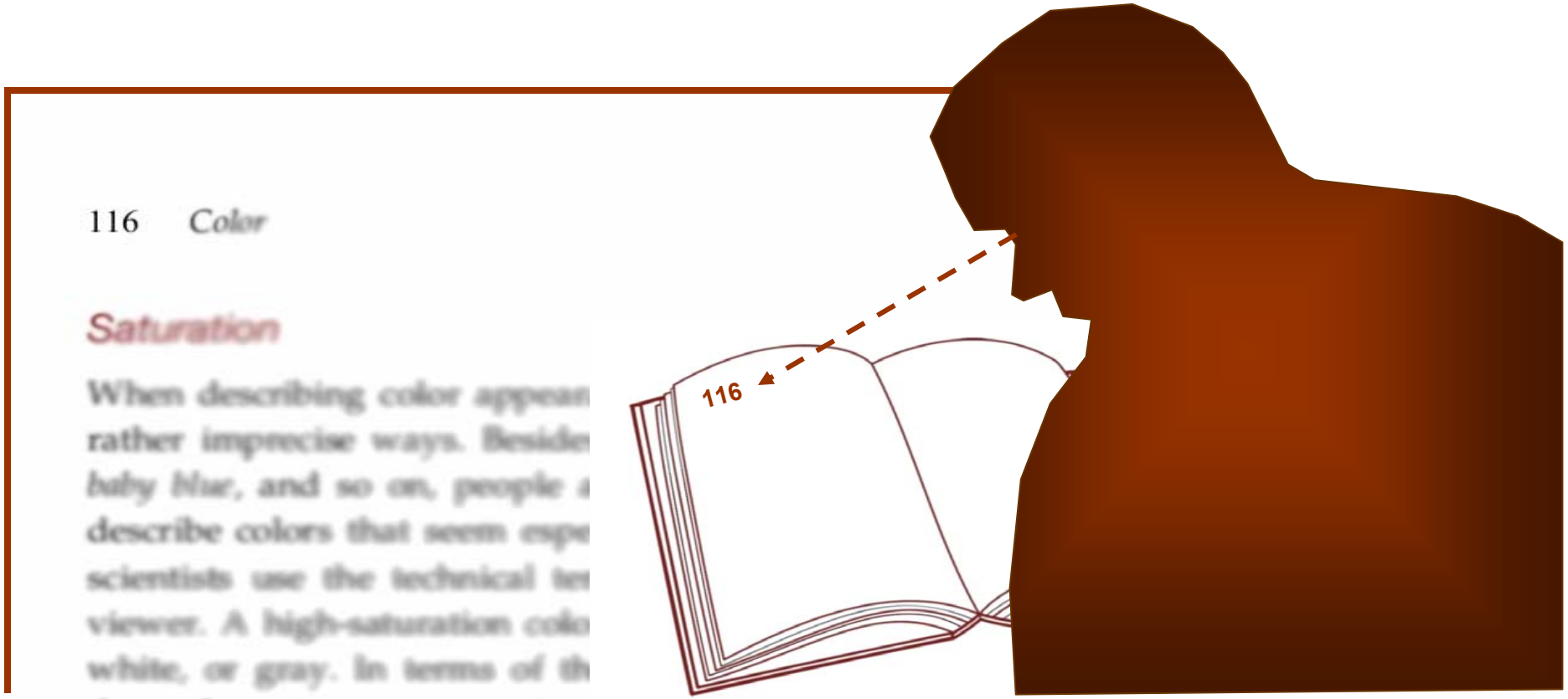




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Need a proof?

- Try reading a book focusing on the page numbers
  - impossible!



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Peripheral vs foveal vision

- Peripheral vision (rods)
  - good at detecting motion
    - a property of rod cells
  - good in the dark
  - weak at distinguishing colors
- Foveal vision (cones)
  - good at detecting details
  - good at distinguishing colors
  - weak in the dark

Our view in the dark



mostly peripheral

mostly black and white

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Averted vision

- Technique for viewing faint objects using peripheral vision
  - https://en.wikipedia.org/wiki/Averted_vision
- It consists of not looking directly at the object, but looking a little off to the side
  - used by astronomers
  - known to Aristotle



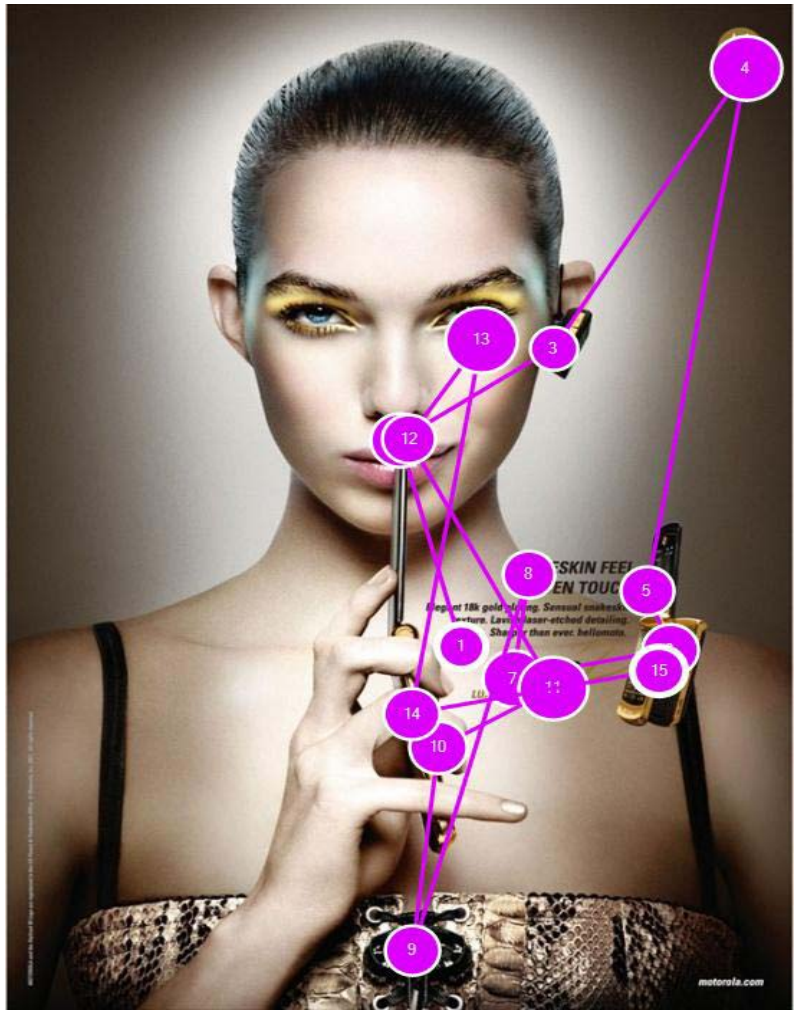
Fixations and saccades

- Saccade (fast eye movement)
  - the muscles accelerate the eyeball to an angular velocity up to 900 degrees per second
    - would allow your eyes to rotate 2.5 times in a second
  - each saccade lasts less than one-tenth of a second
  - our eyes perform up to 5 saccades per second

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Eyetracking example





From: http://eyetracking.com.ua

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Vision suppression

- During saccadic movement vision is suppressed
  - extremely long saccades may allow you to perceive a black flash in the middle
    - try yourself: without moving your head and with your arms spread open, watch alternatively your left and right hand

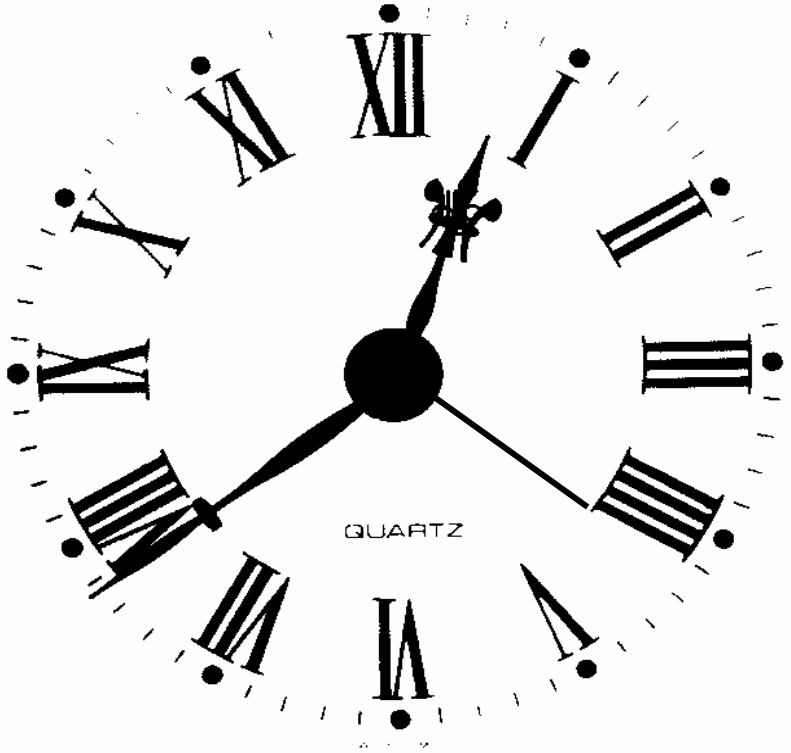


Long saccade
(and black flash in the middle)

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Clock illusion

- Quickly checking your watch, at first the arm of the seconds seems not to move
  - you need an analogic clock with the arm for the seconds in continous movement
  - best perceived after a long saccade



- Your brain strives to reconstruct a continous scene from discrete snapshots

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Seeing is a complex task



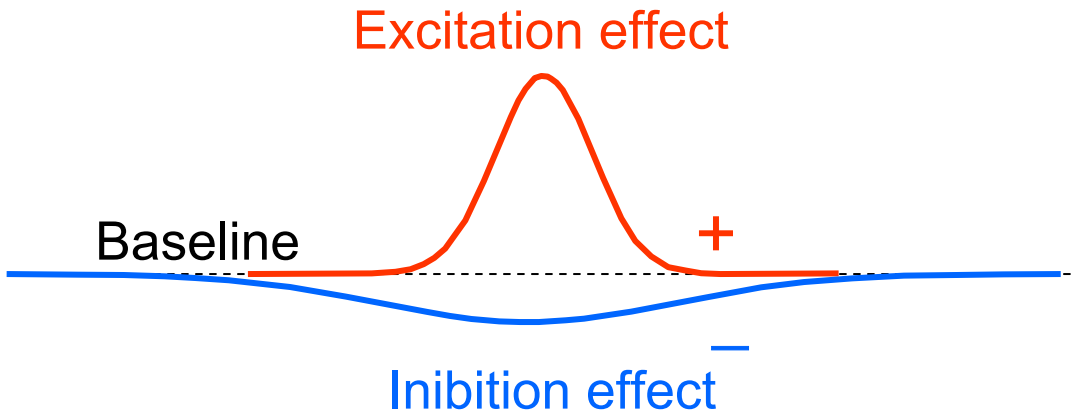
- Vision is an active reconstruction process
- quick eye movement provide a sequence of discrete snapshots
  - detailed fragments from high-resolution fovea
  - B&W background images and motion perception from low-resolution peripheral view

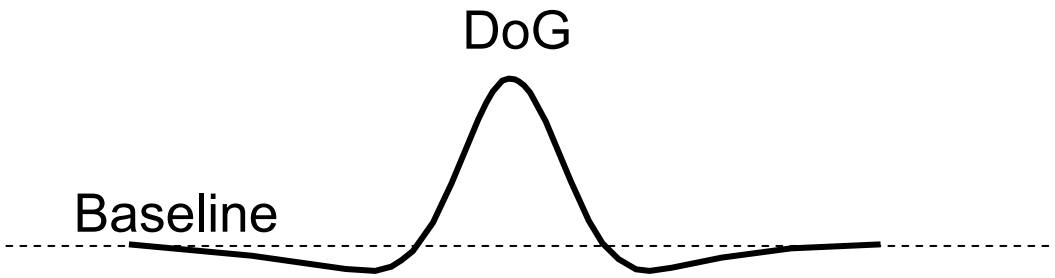
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DoG Function



- DoG = Difference of Gaussians
- A photon hitting a receptor contributes with two different effects
  - strong and local excitation effect
  - weak and spread inibition effect
- The sum of the two effects (DoG) yields an edge detection filter



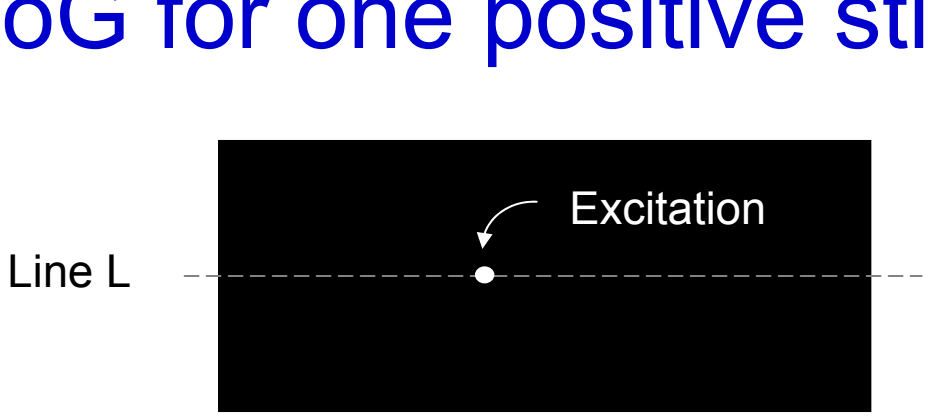


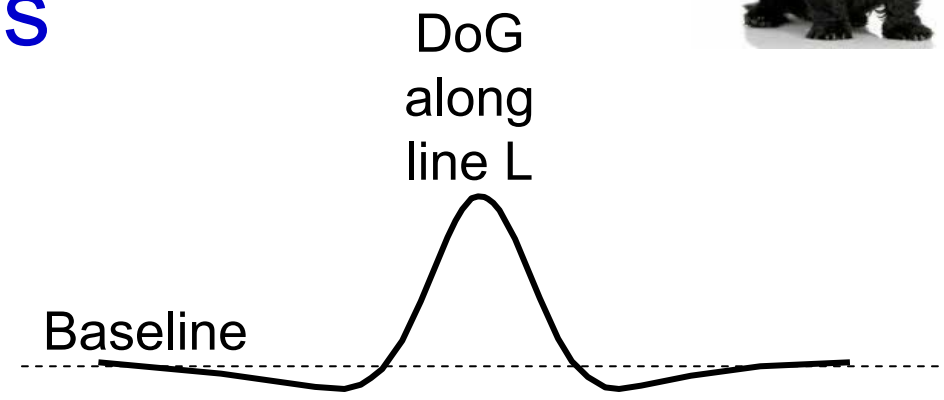
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DoG Function

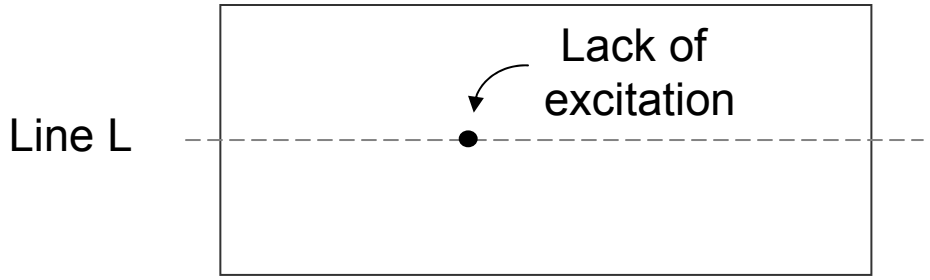


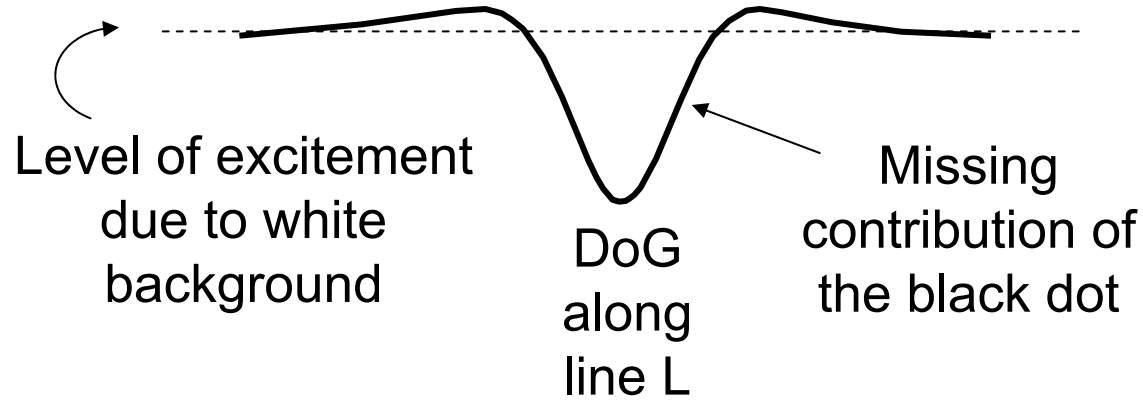
- DoG for one positive stimulus





- You may also think of DoG in terms of negative stimula (lack of excitations)





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Edge detection with DoG

- Which is brighter?





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Edge detection with DoG

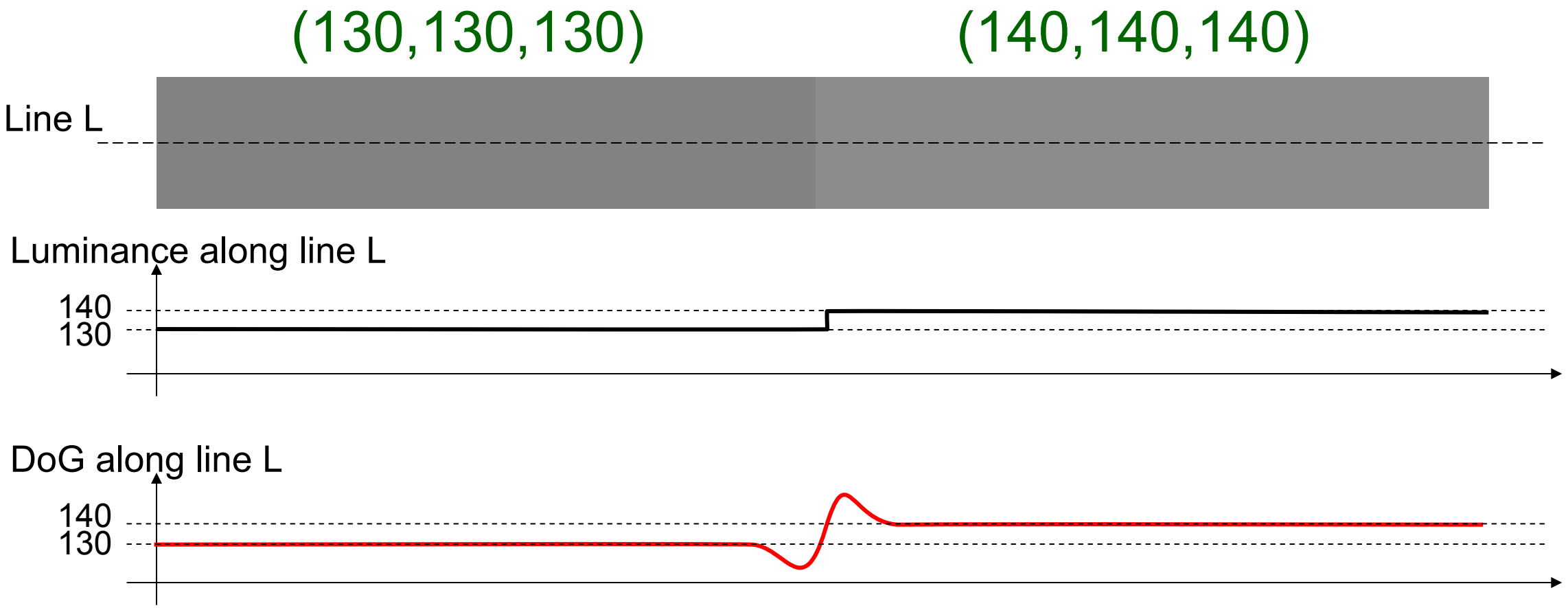
- Which is brighter?



(130,130,130) (140,140,140)

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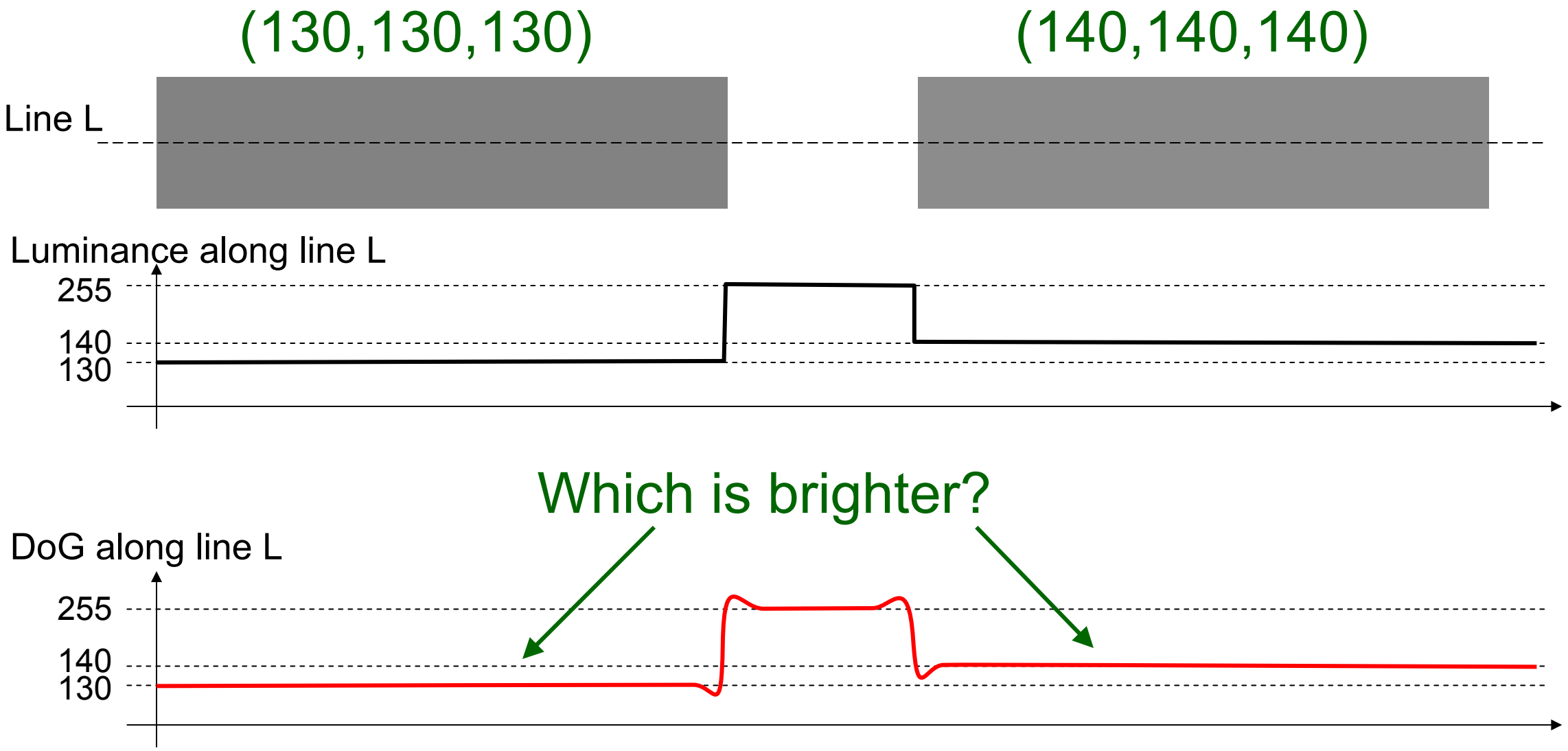
Edge detection with DoG



- The cumulative effect of DoG is that of detecting edges

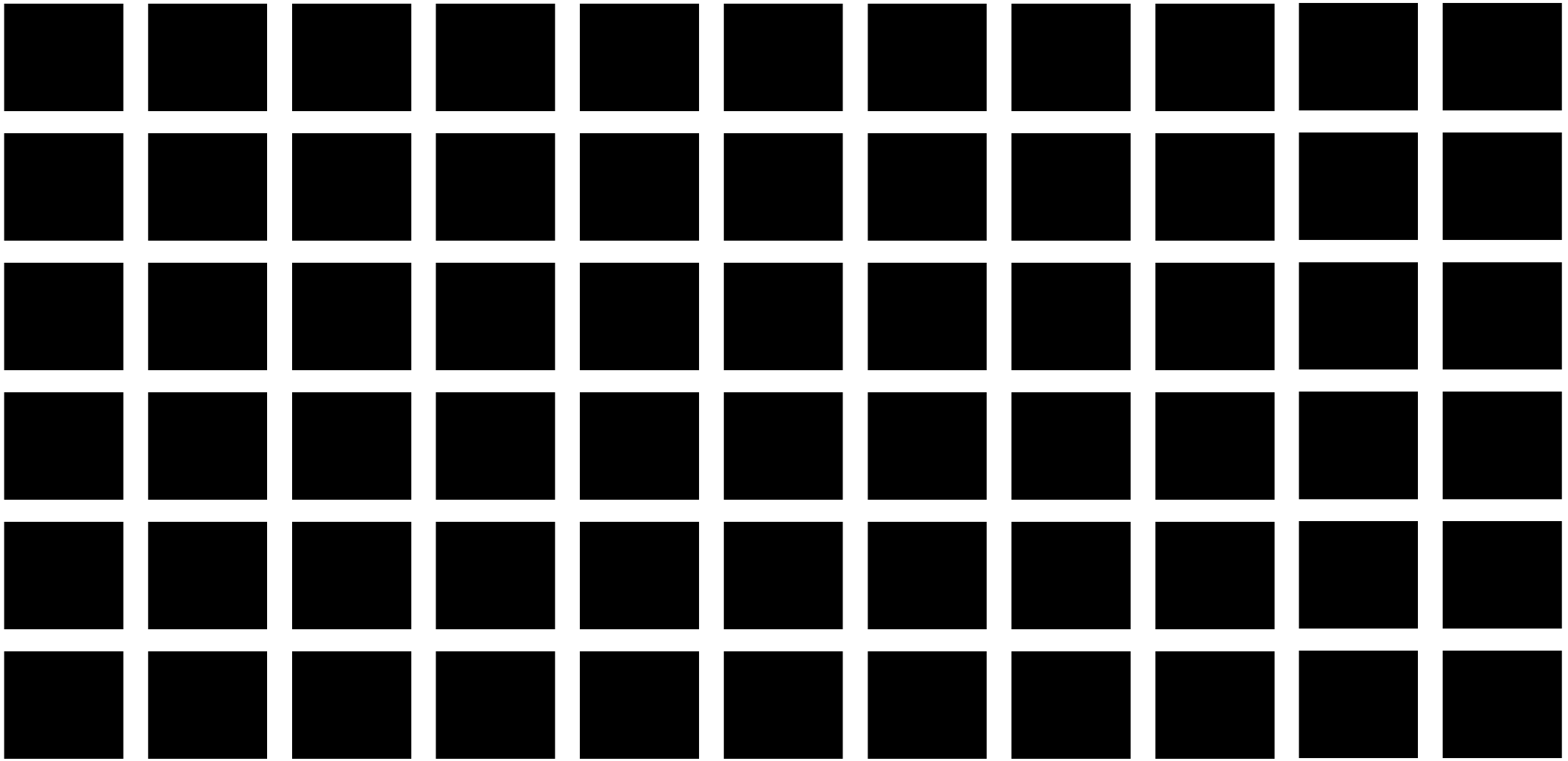
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Edge detection with DoG



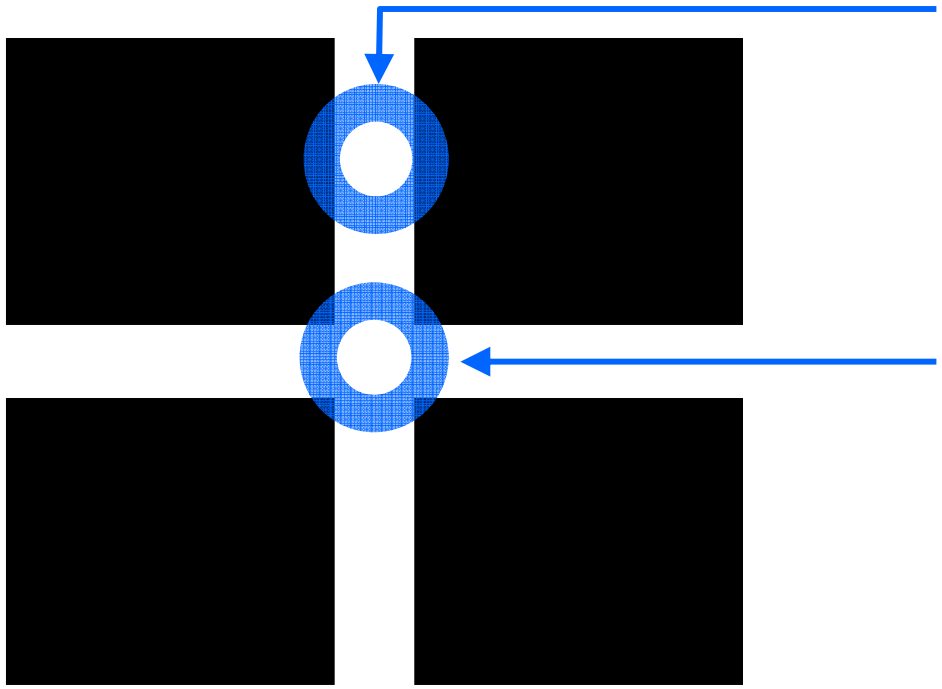
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Hermann grid illusion



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Hermann grid illusion explained



More black in the neighborhood

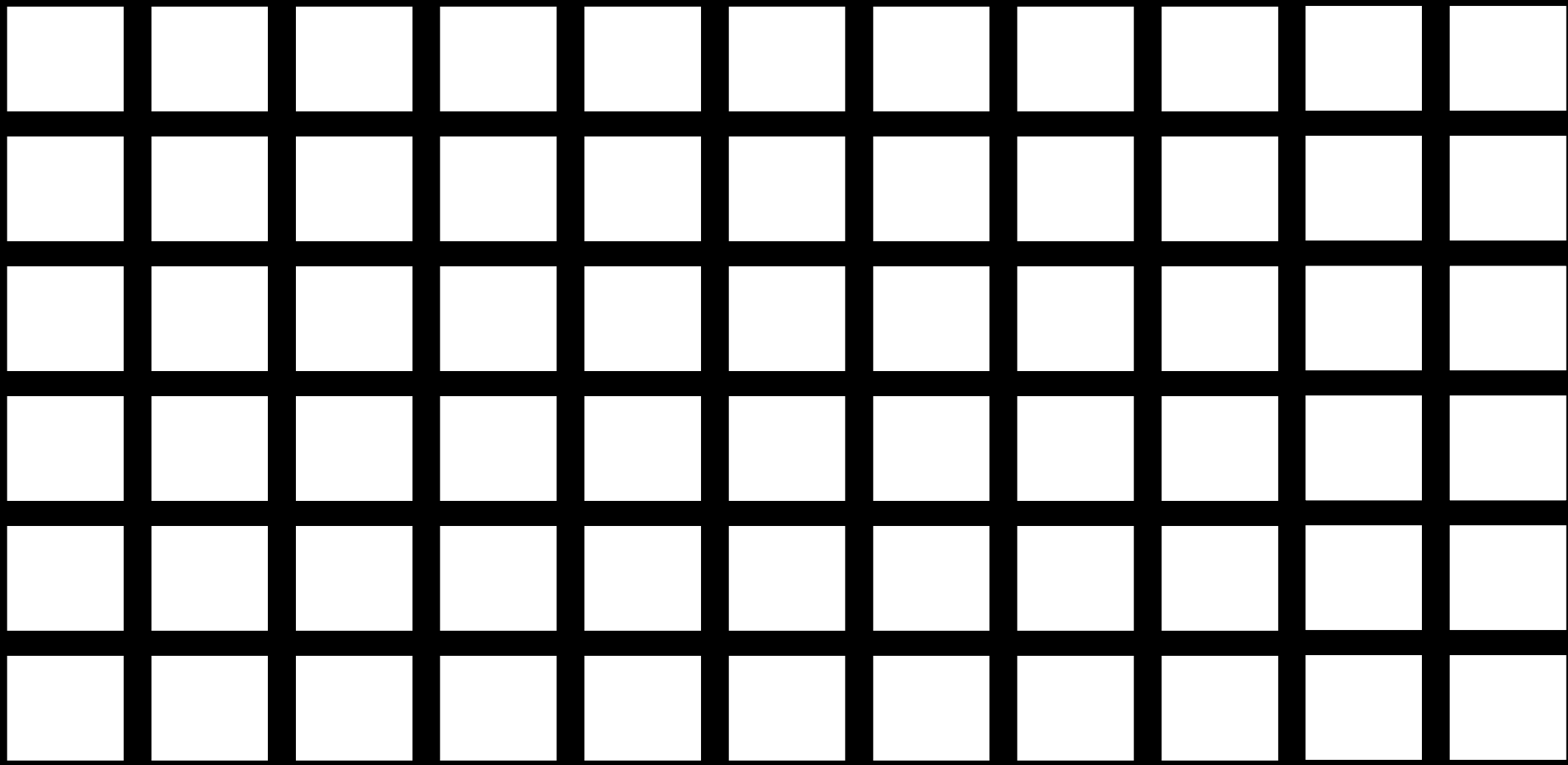
- white patch is perceived brighter

Less black in the neighborhood

- white patch is perceived darker

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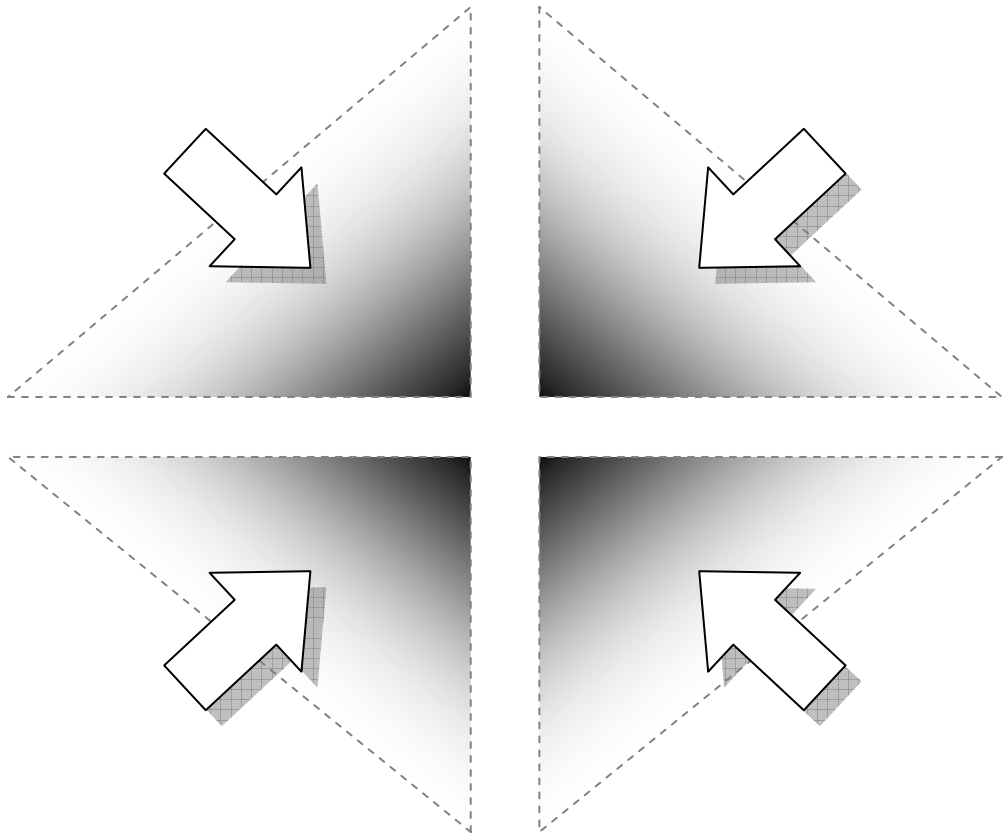
Hermann grid illusion (negative)

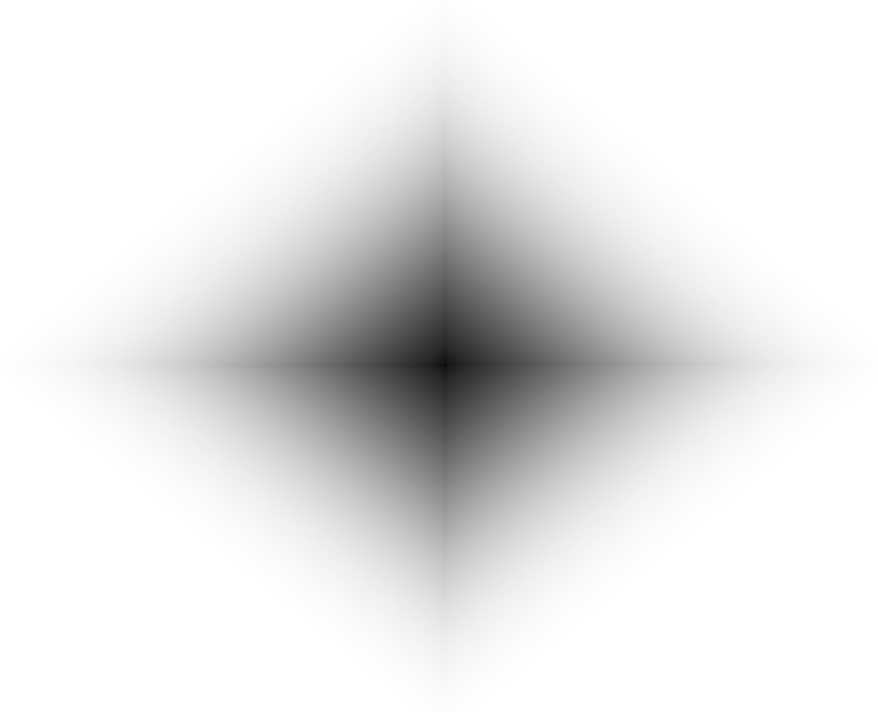


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Mach bands

- Four shaded triangles are joined together
- A darker cross seems to be in the middle

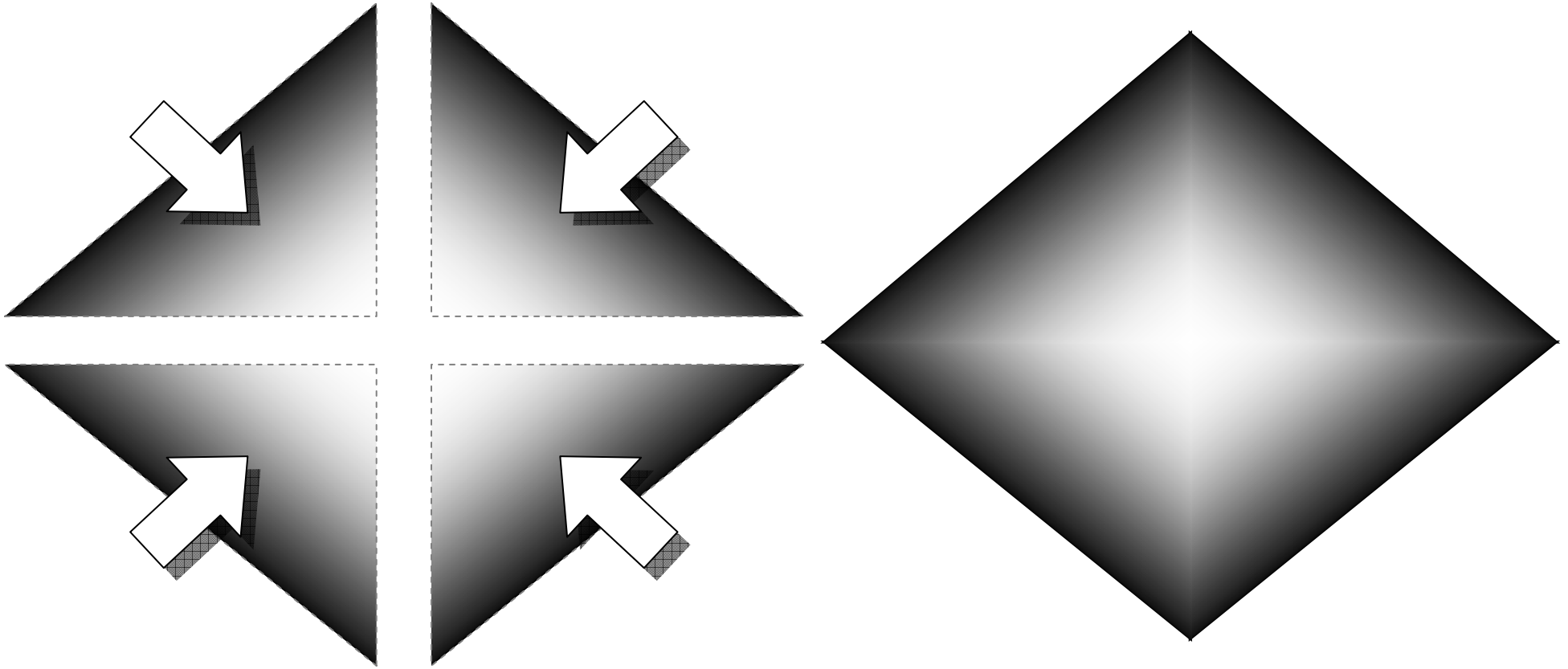




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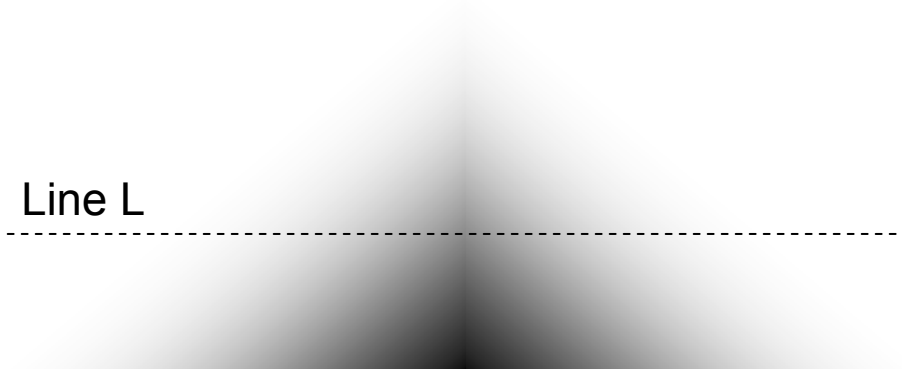
Mach bands

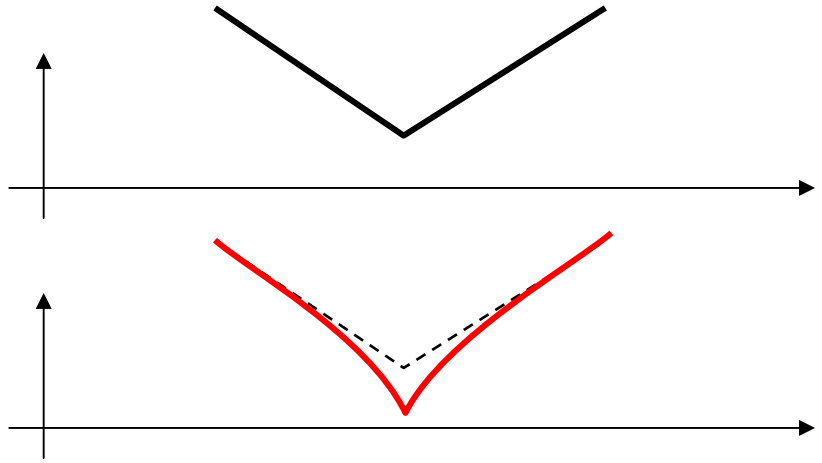
- The negative version also works

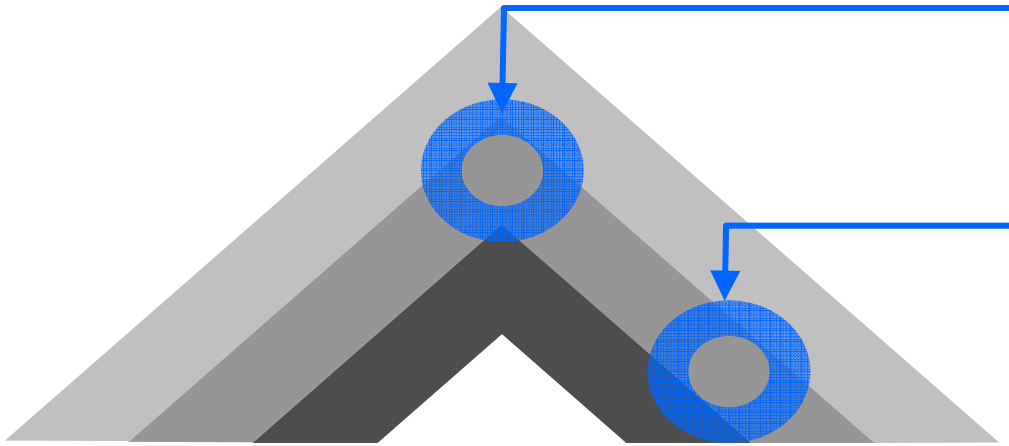


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Mach bands explained







- Brighter neighborhood
  - this patch is perceived darker
- Darker neighborhood
  - this patch is perceived brighter

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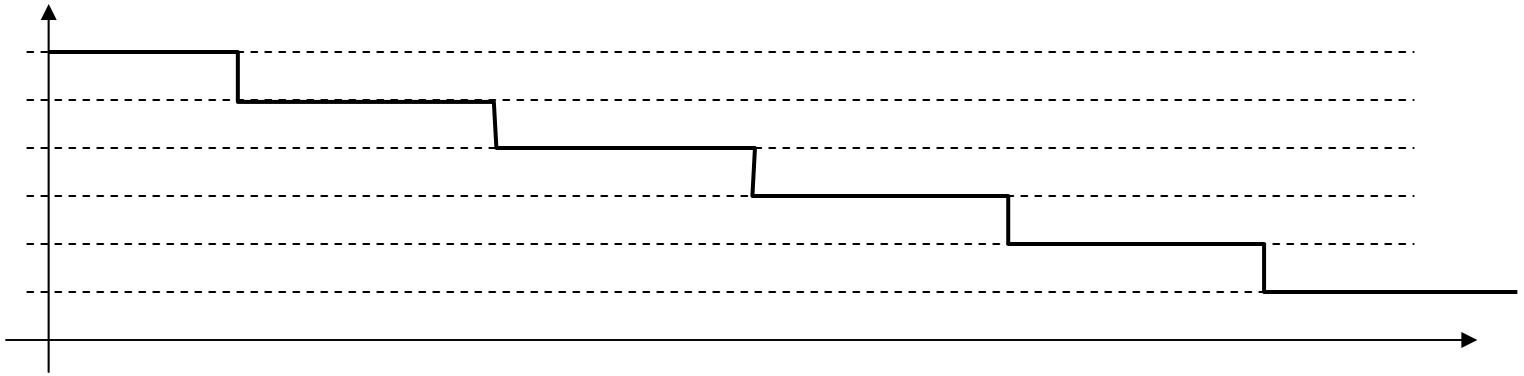
Chevreul illusion



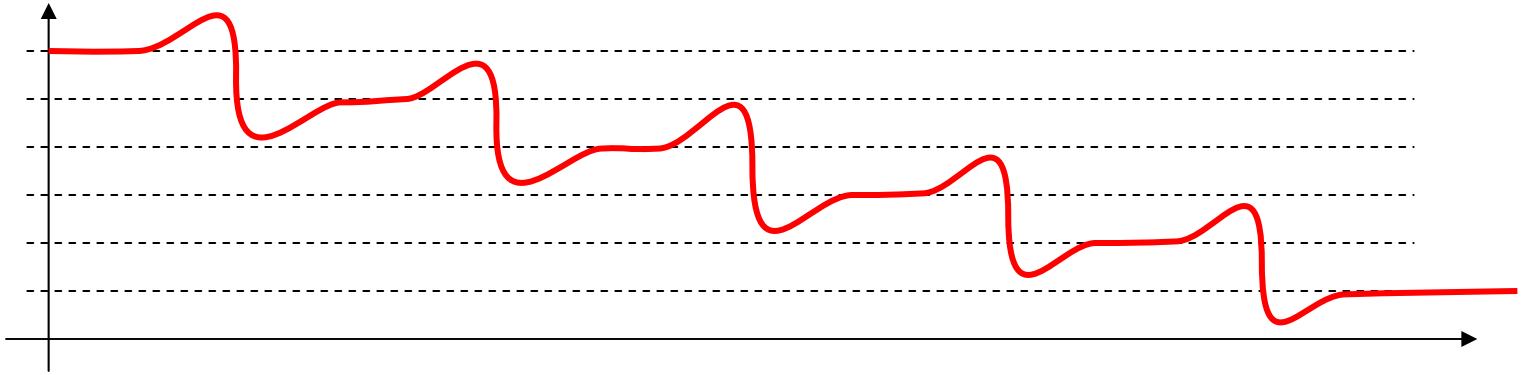
Chevreul illusion explained

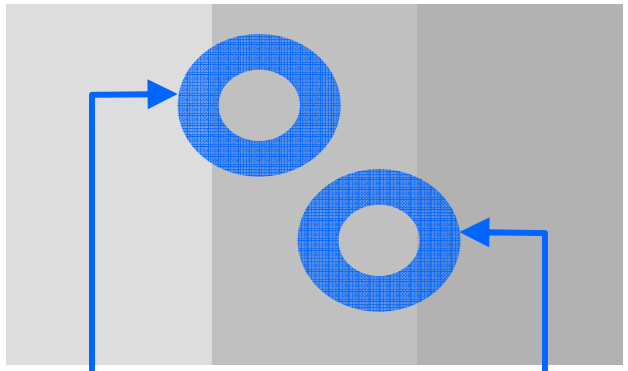


Luminance



DoG



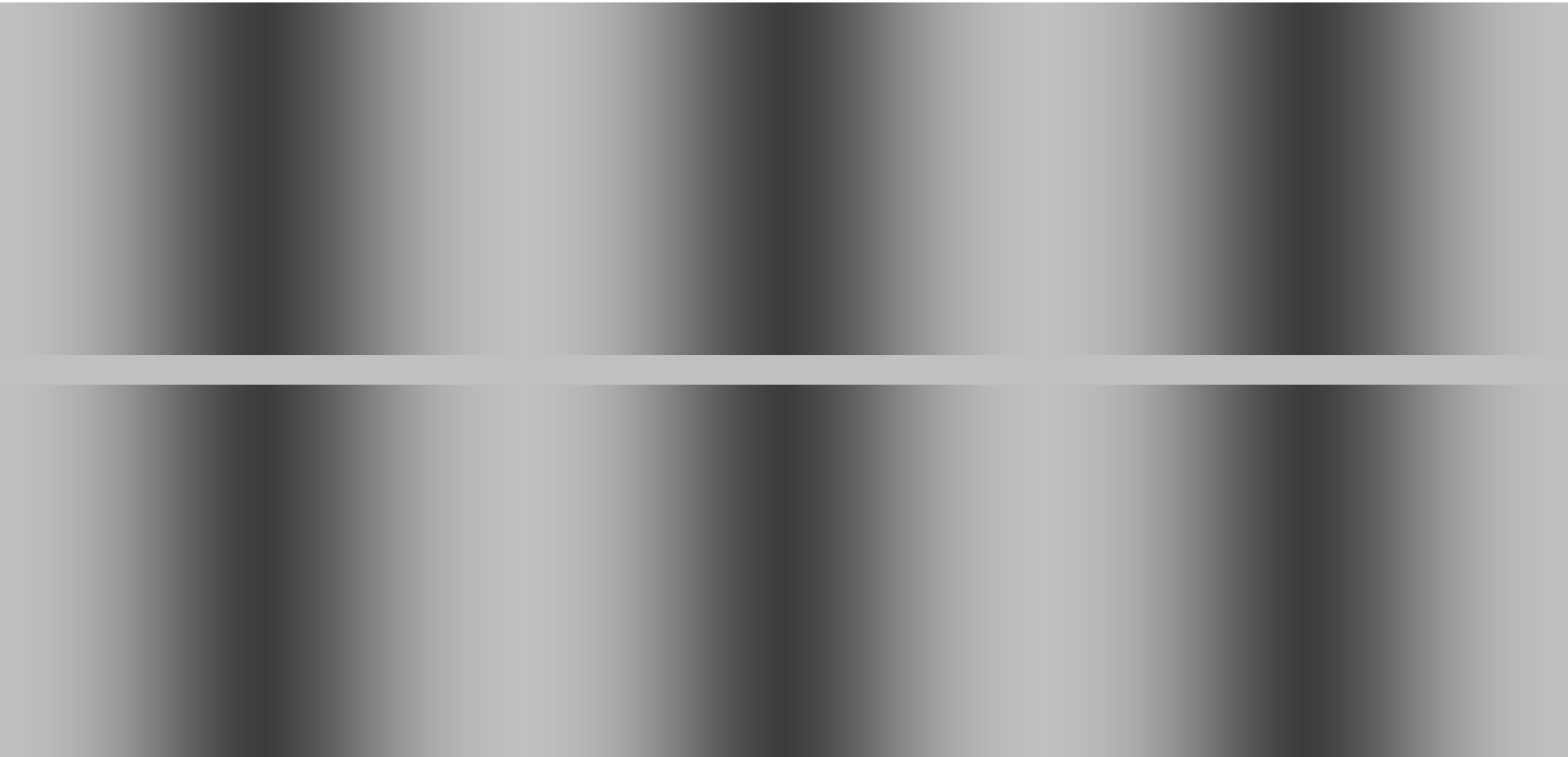


Perceived darker

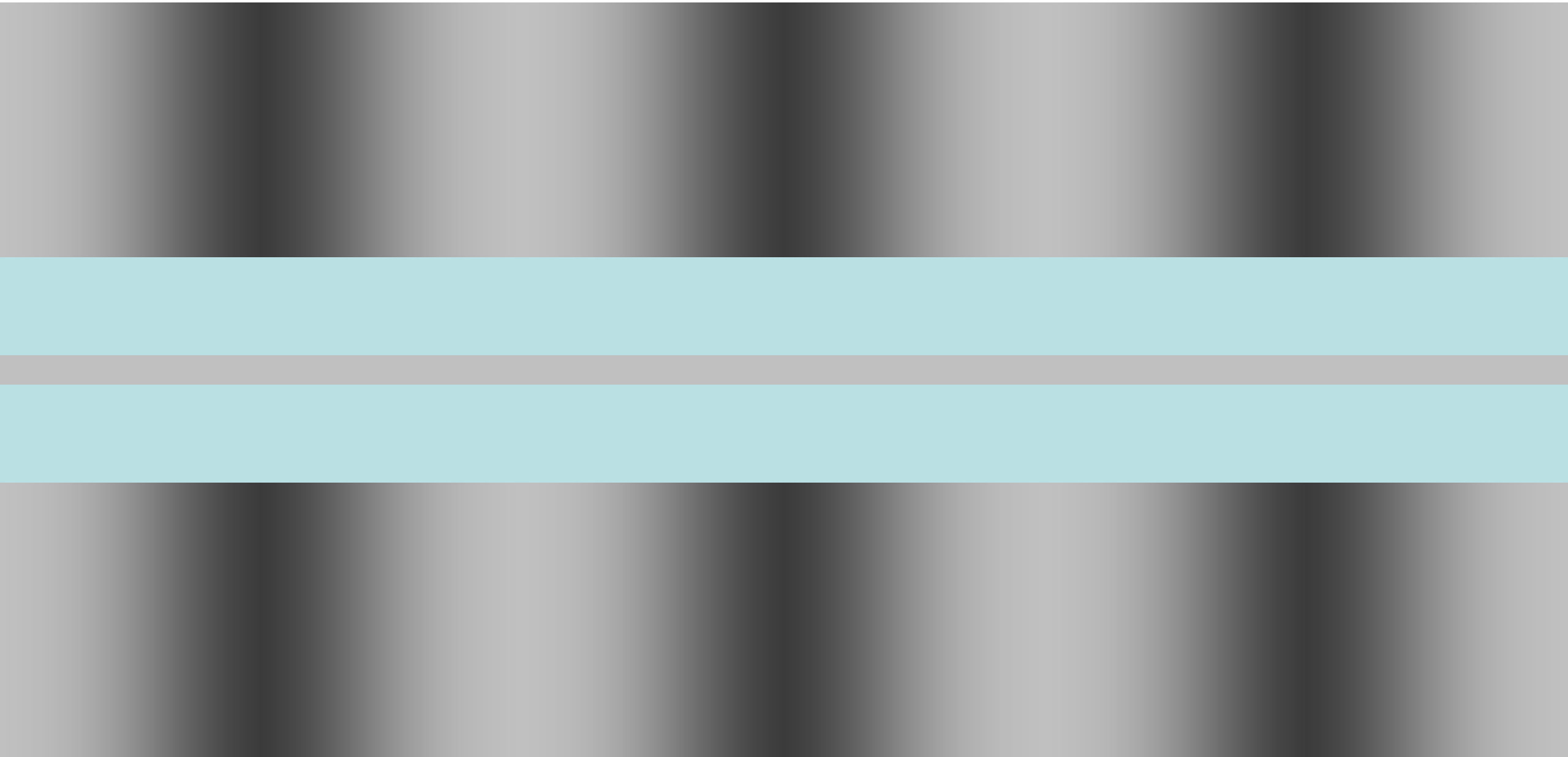
Perceived brighter

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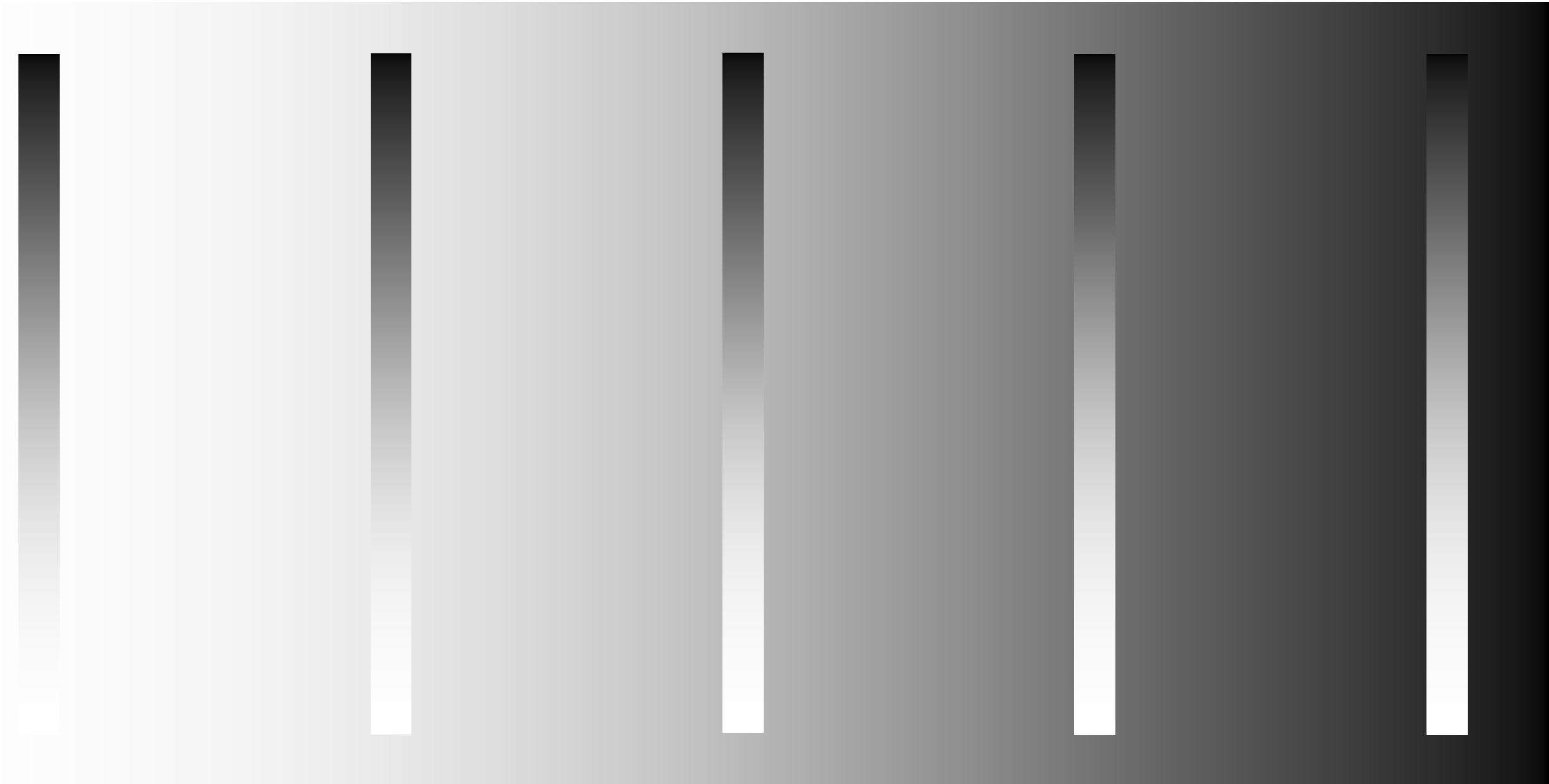
Induced Grating



Induced Grating

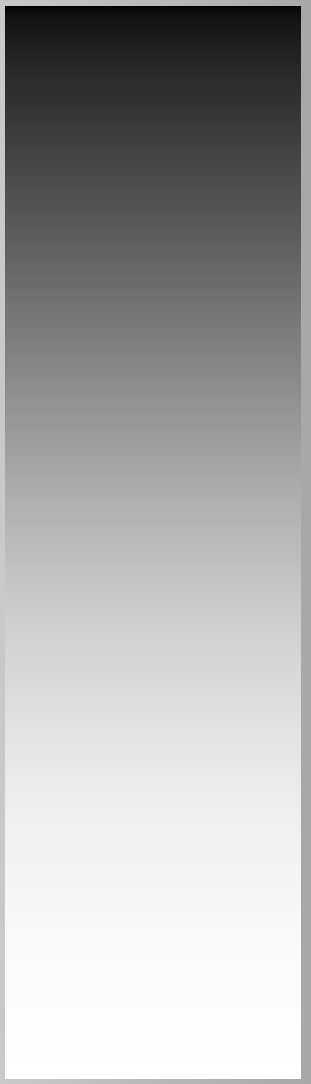


Crispening effect



Crispening effect

- The scales appear to divide perceptually at the value of the background



Tricking DoG

- Which is brigther?



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Tricking DoG

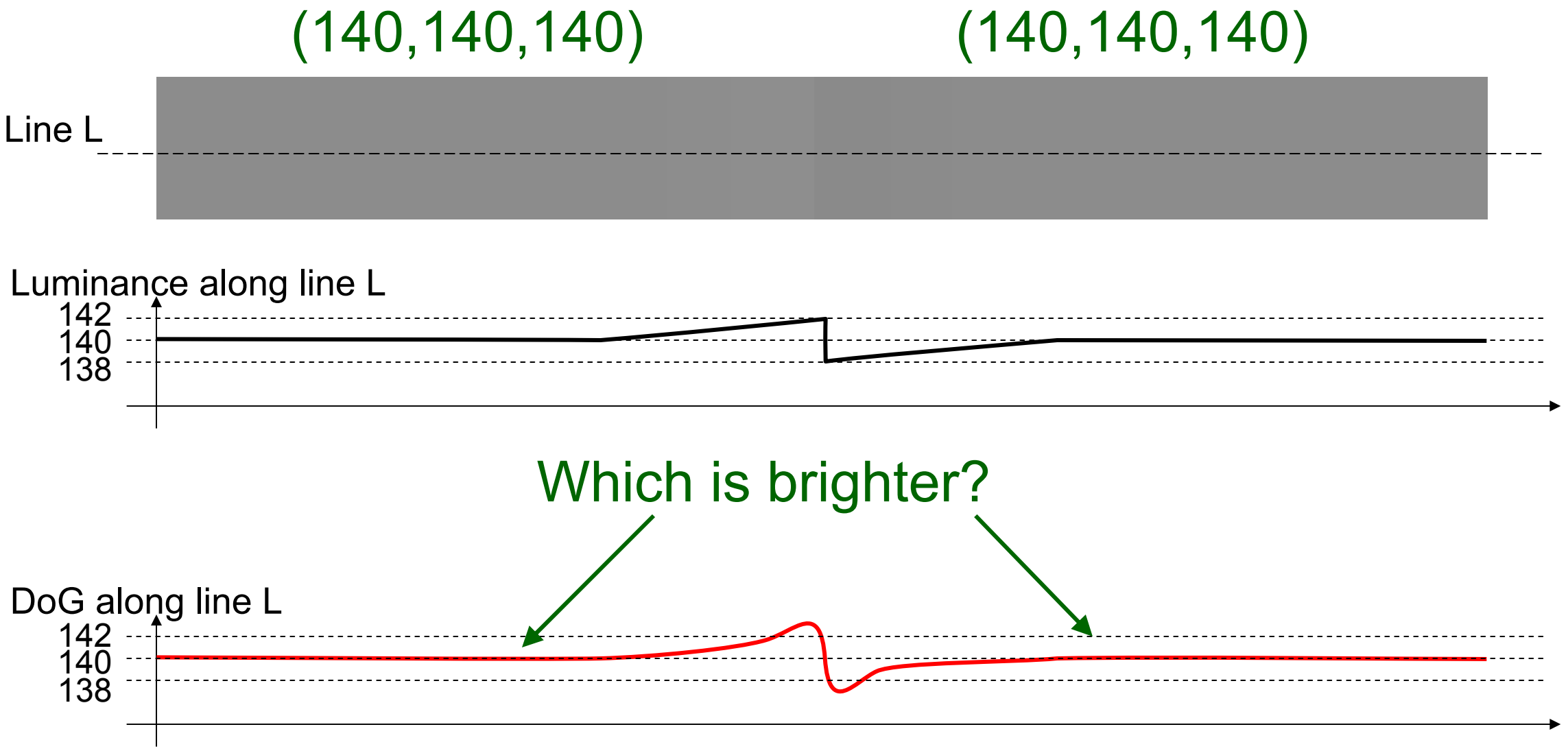
- Which is brigther?



(140,140,140) (140,140,140)

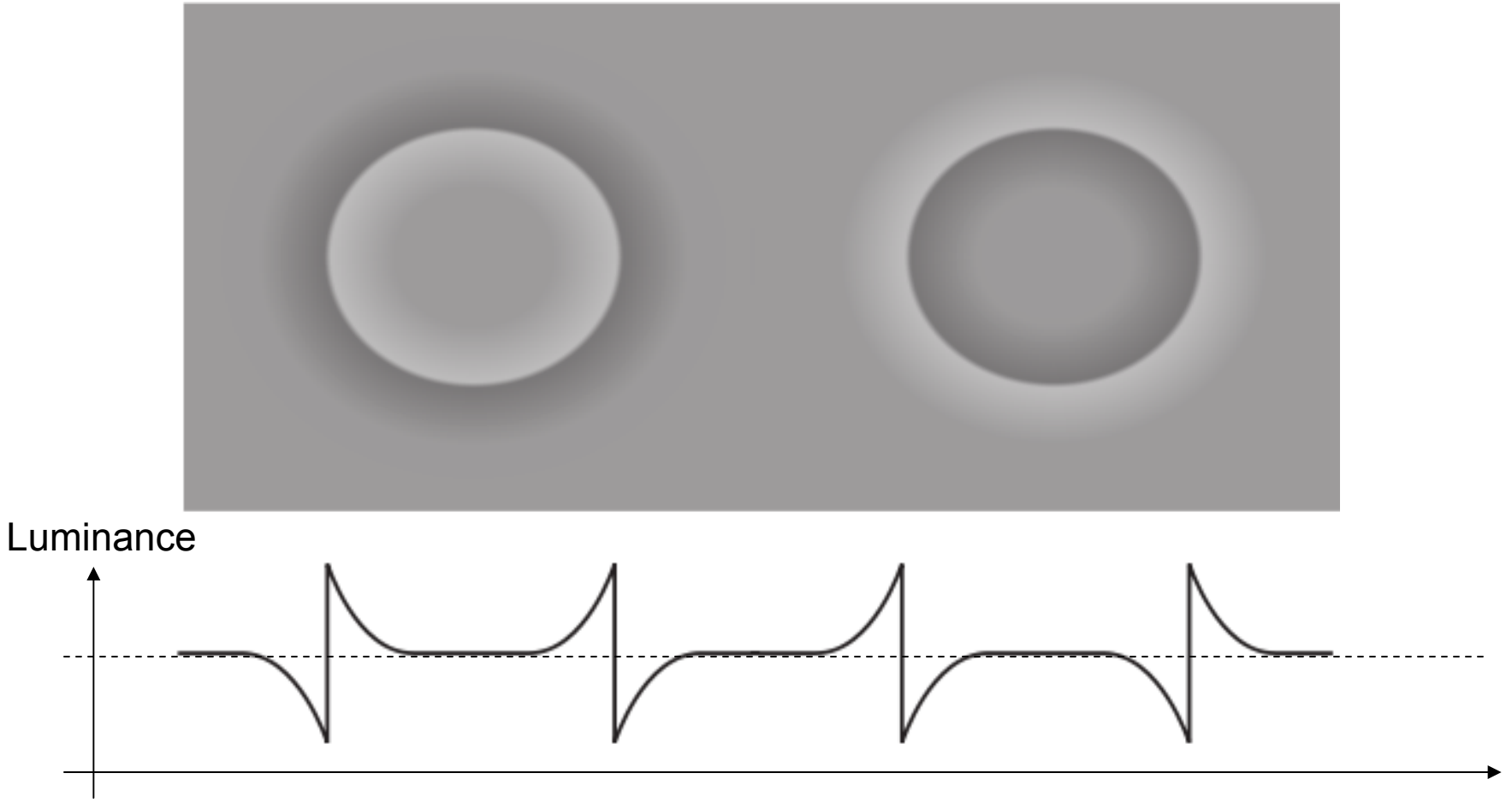
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Tricking DoG



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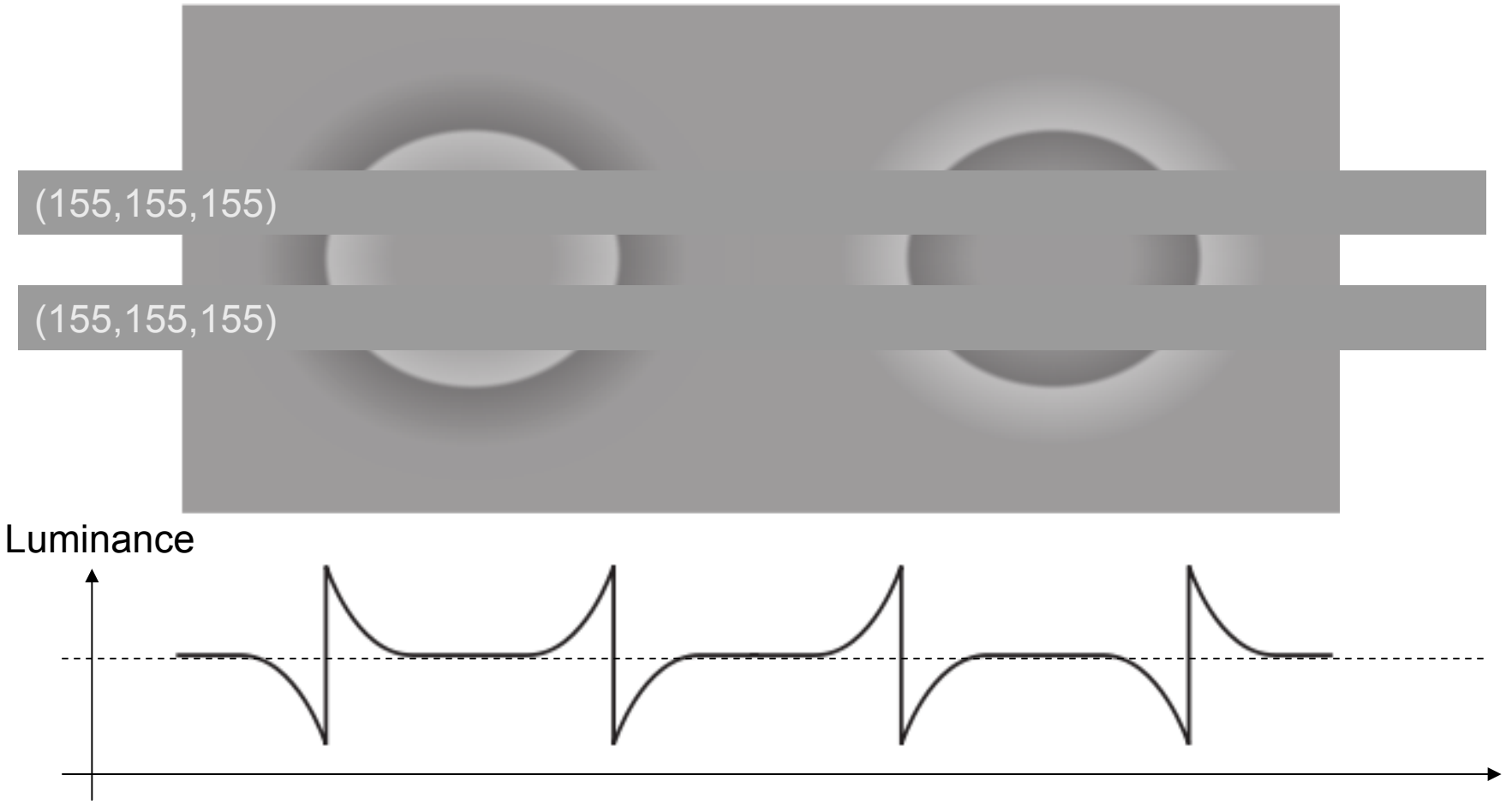
Tricking DoG: Cornsweet effect



- From [Ware 13]

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Tricking DoG: Cornsweet effect

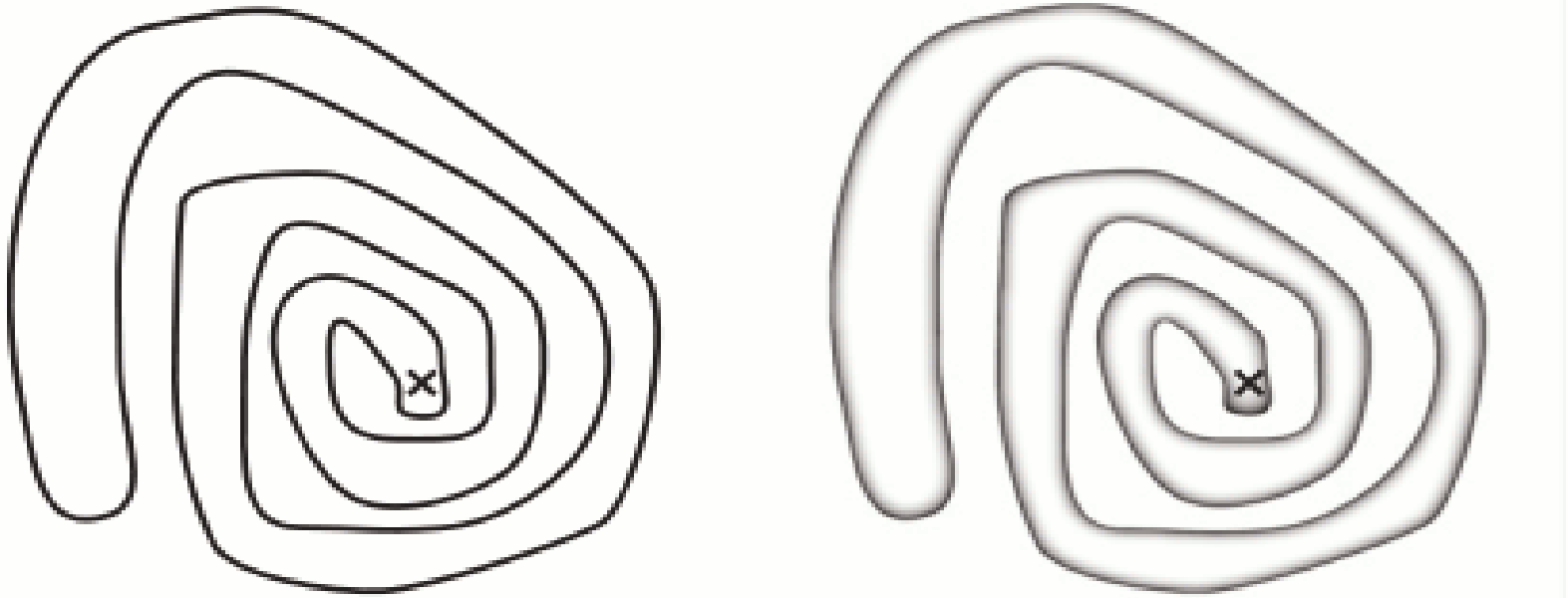


- From [Ware 13]

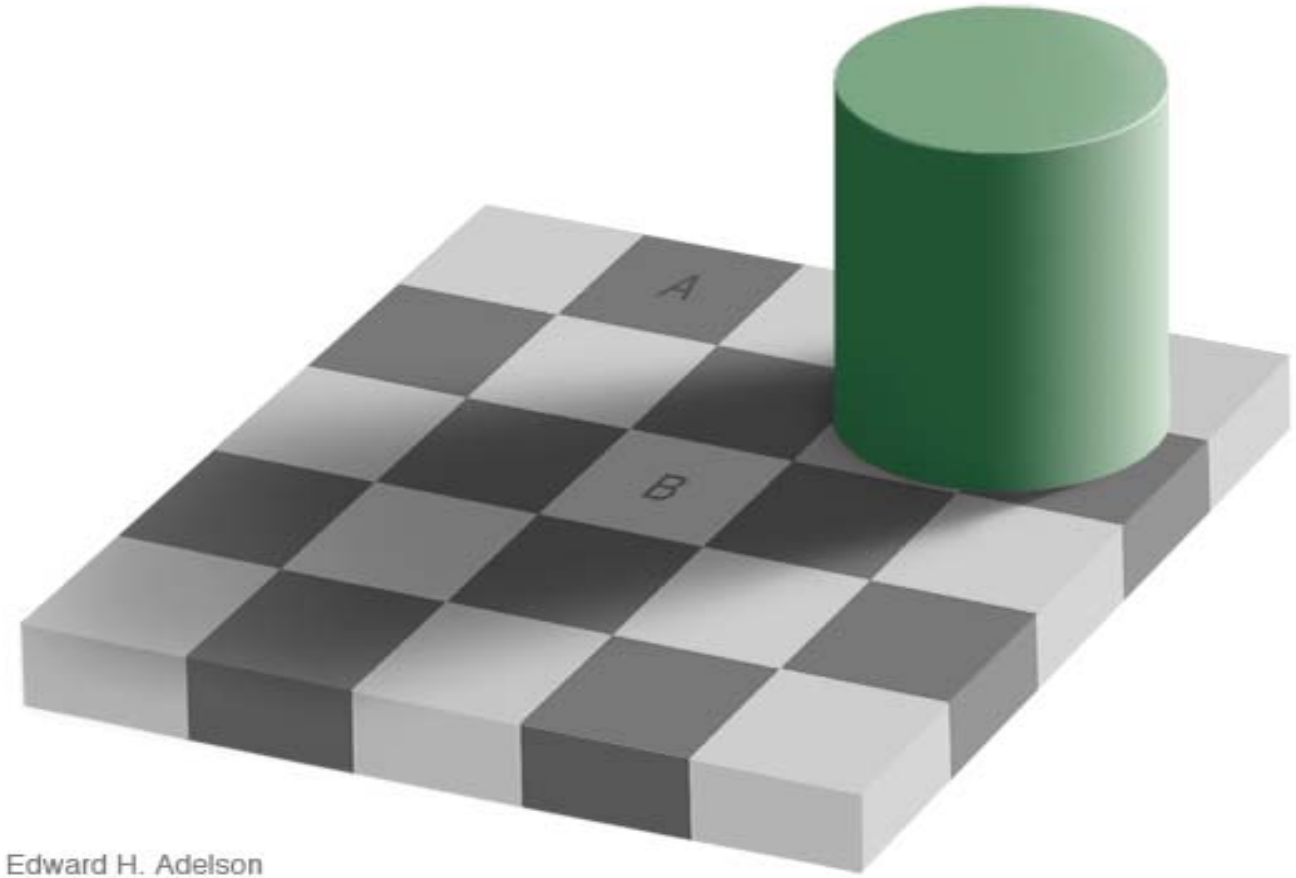
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Cornsweet contours

- It is difficult to see if the *x* is inside or outside of the bounded region
- Using Cornsweet contours makes it possible to see the solution much more rapidly
  - From [Ware 13]



Checkershadow illusion

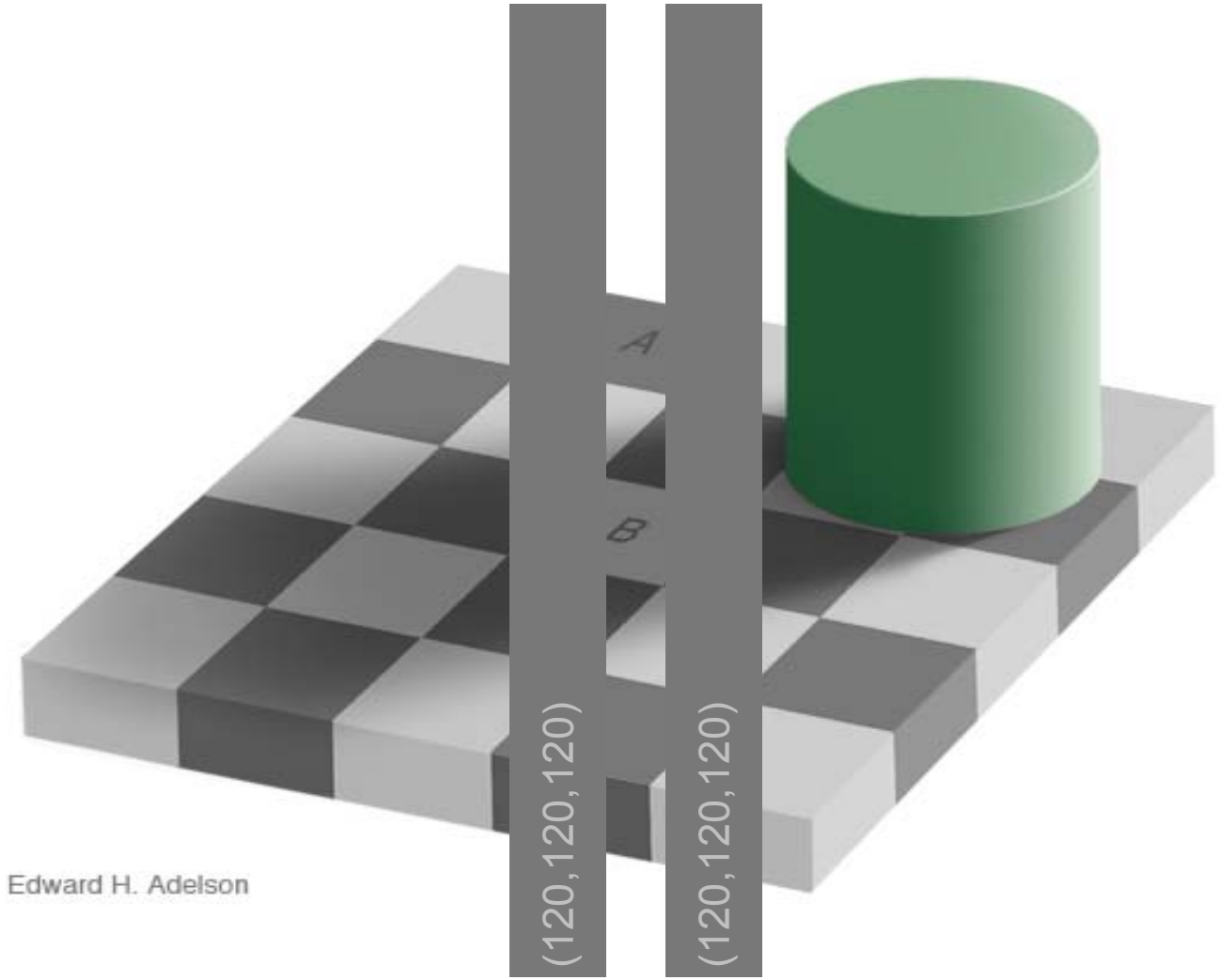


Edward H. Adelson

- Lightness perception is based on relative, not absolute, judgements
  - from [Adelson ‘95]

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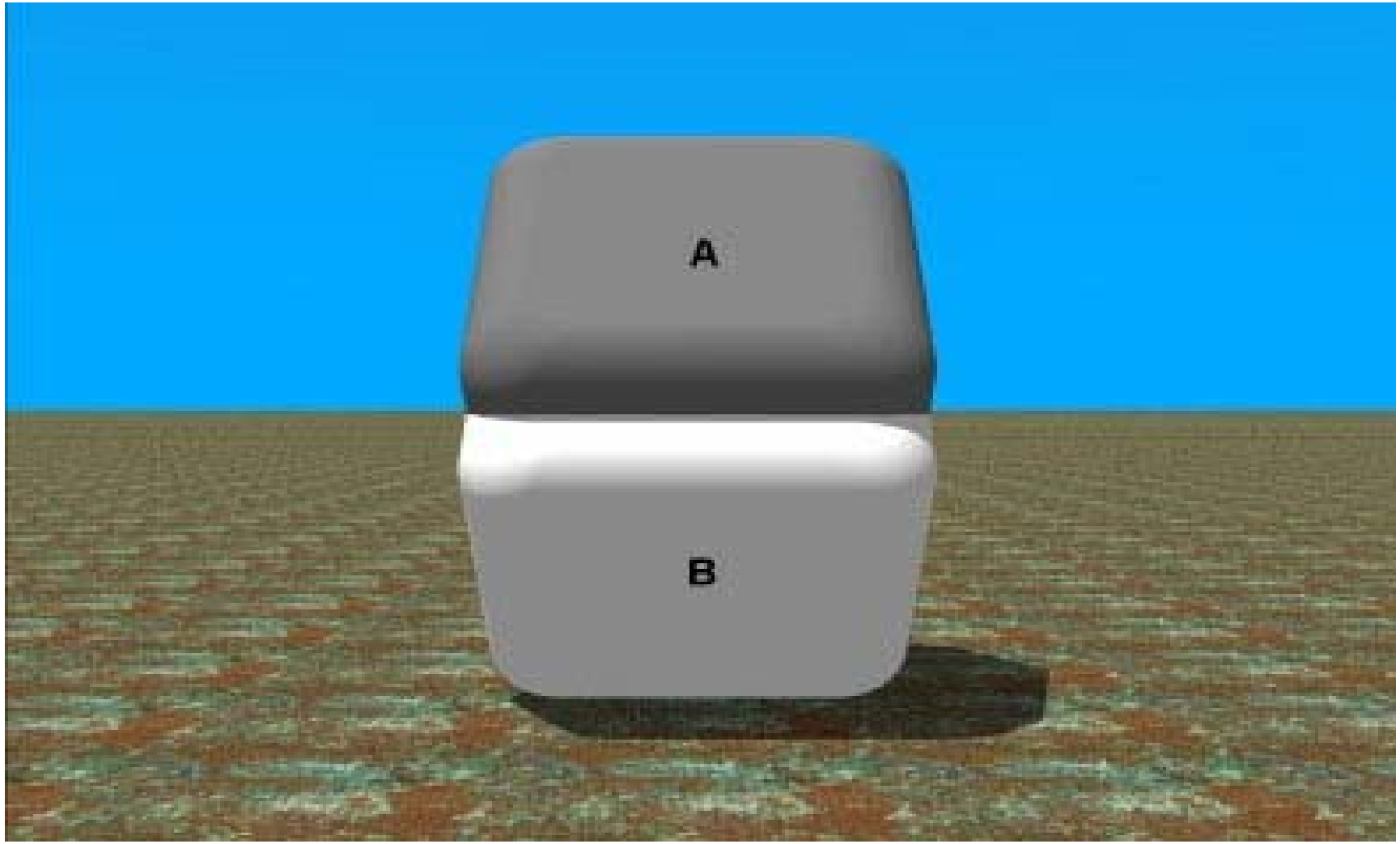
Checkershadow illusion



- Lightness perception is based on relative, not absolute, judgements
  - from [Adelson '95]

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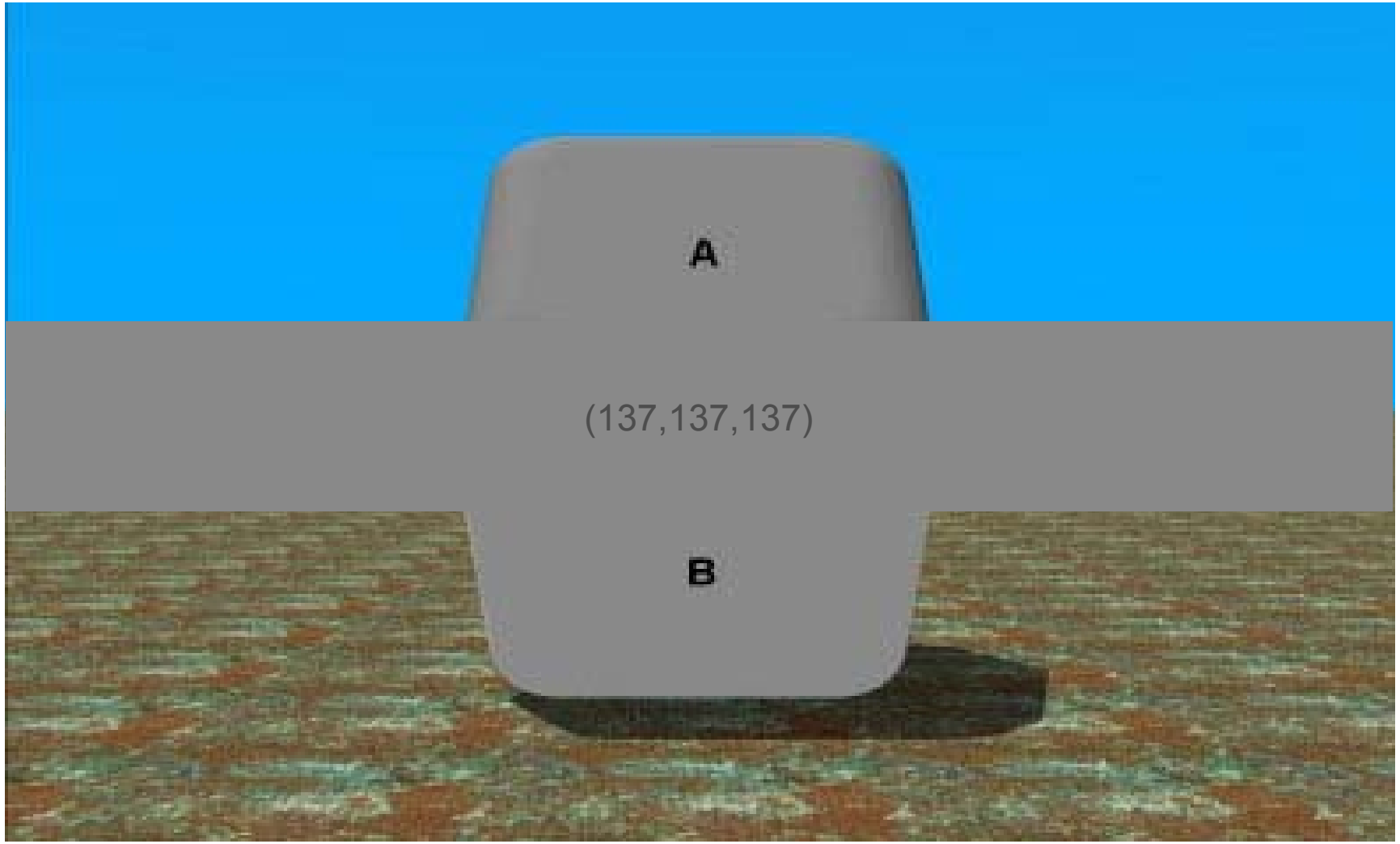
Lightness perception illusion



- Cube illusion
  - [Lotto 14]

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Lightness perception illusion

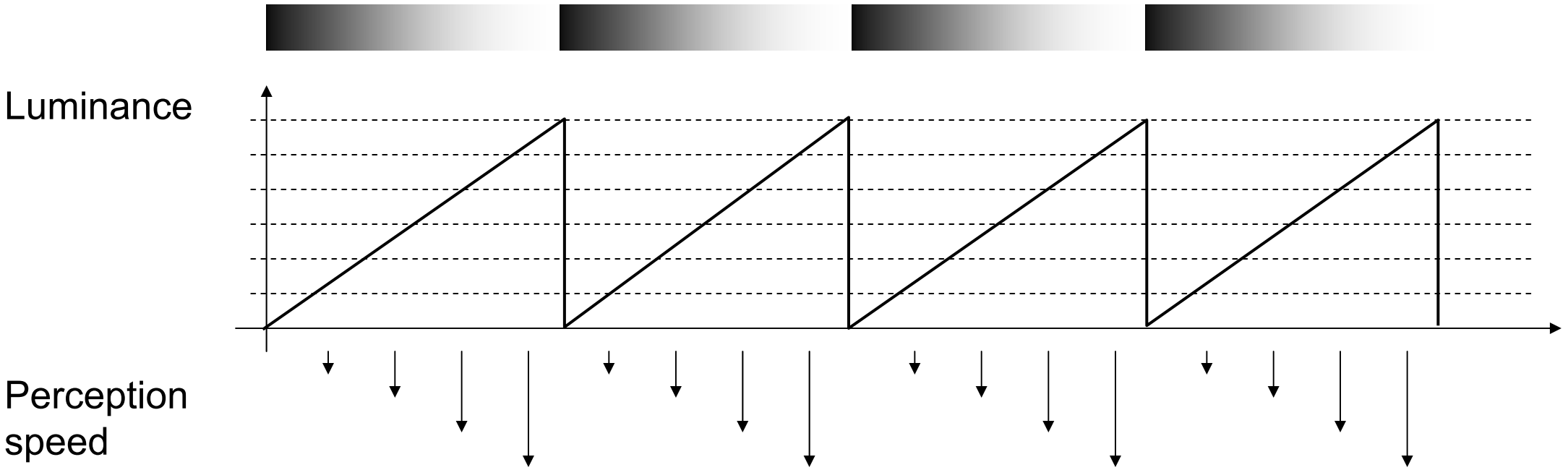


- Cube illusion
  - [Lotto 14]

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Peripheral drift illusion

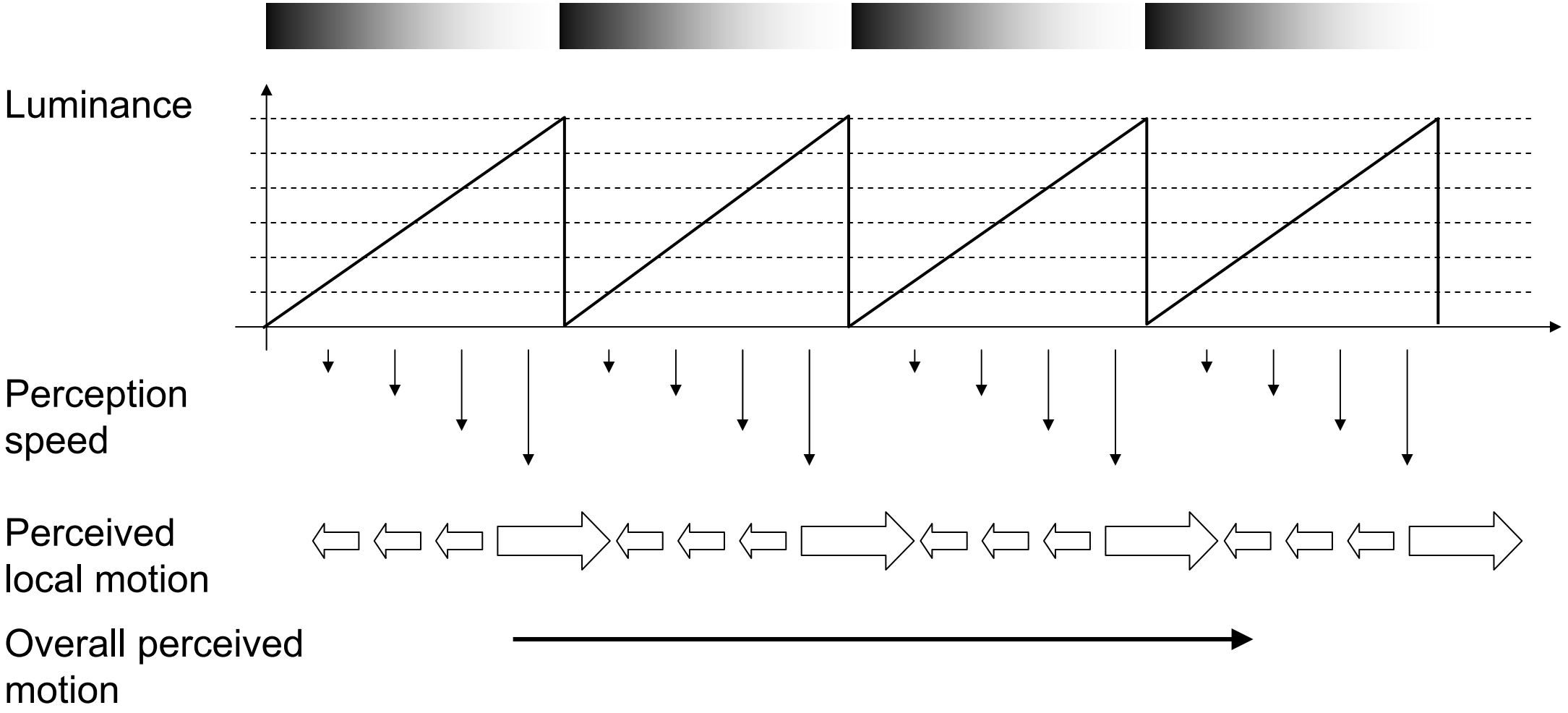
- Motion illusion only affecting the peripheral view
- Luminance information travels through the visual system at different latencies
  - lighter information passes through the system faster than darker information



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Peripheral drift illusion

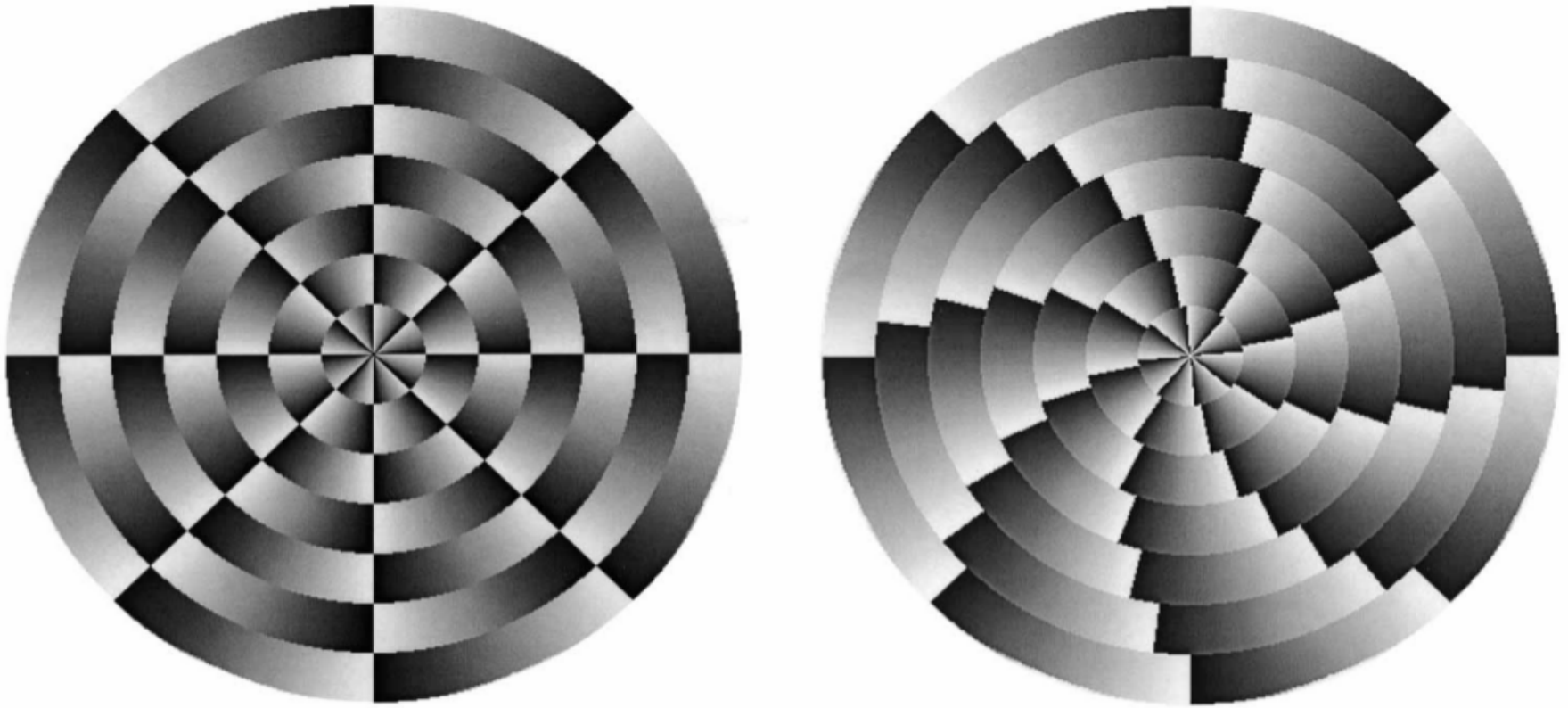
- Stronger perception delays are erroneously interpreted as motion [Faubert & Herbert 99]



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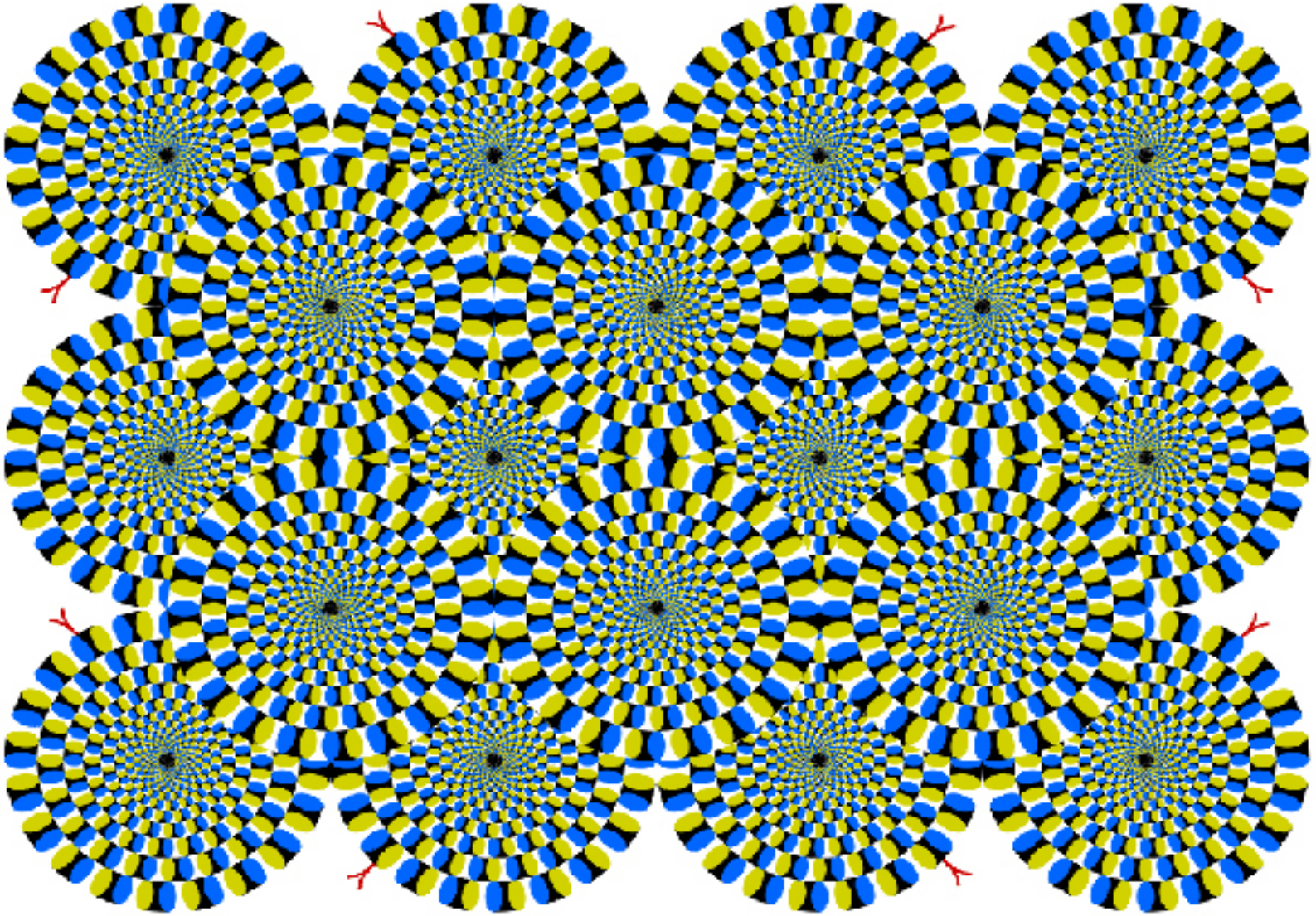
Examples of motion illusions

- From [Faubert & Herbert 99]



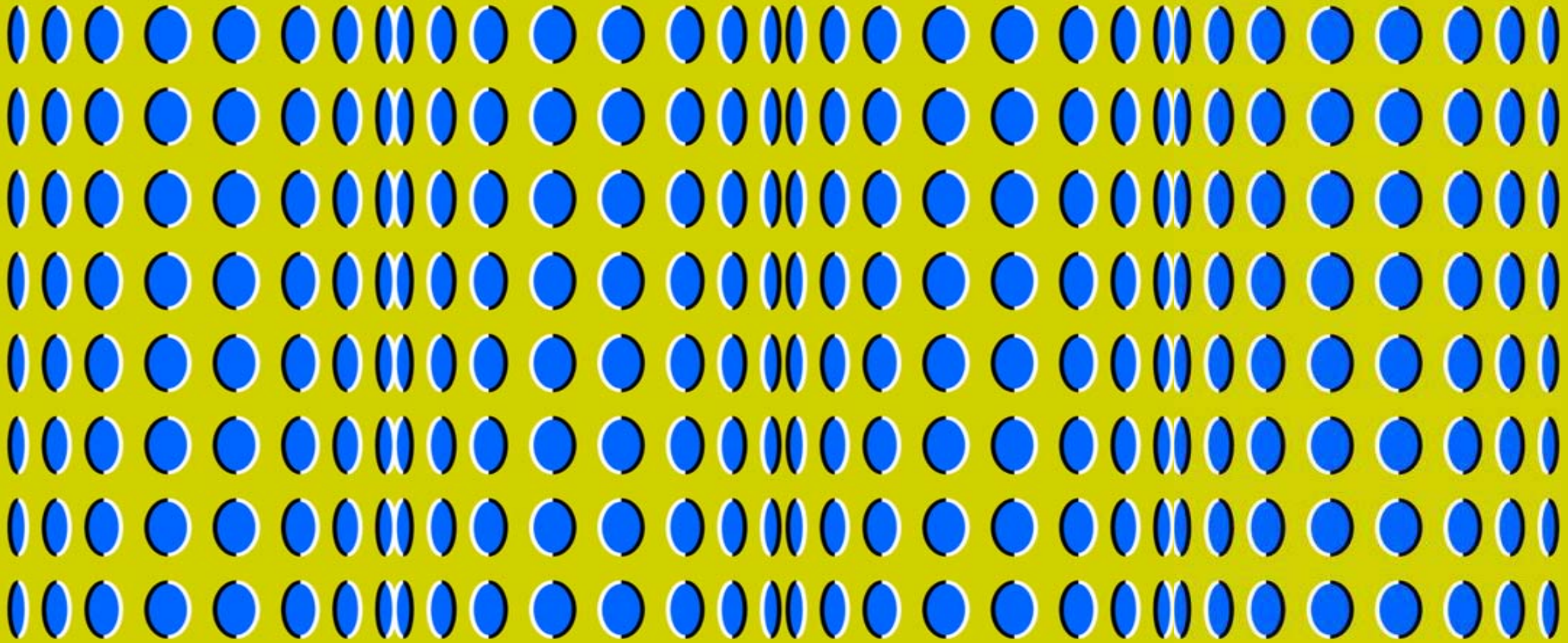
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Examples of motion illusions



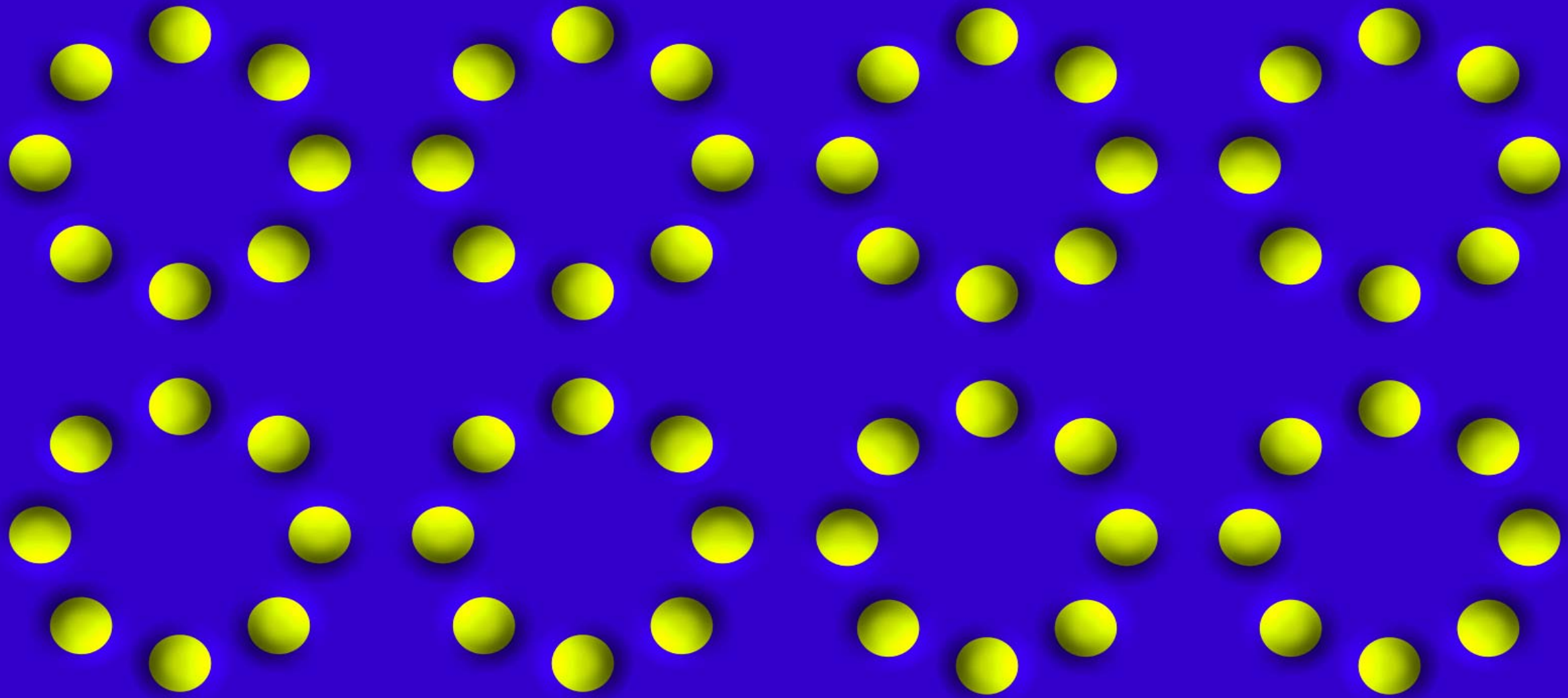
“Rotating snakes” by Akiyoshi Kitaoka [Kitaoka 18]

Examples of motion illusions



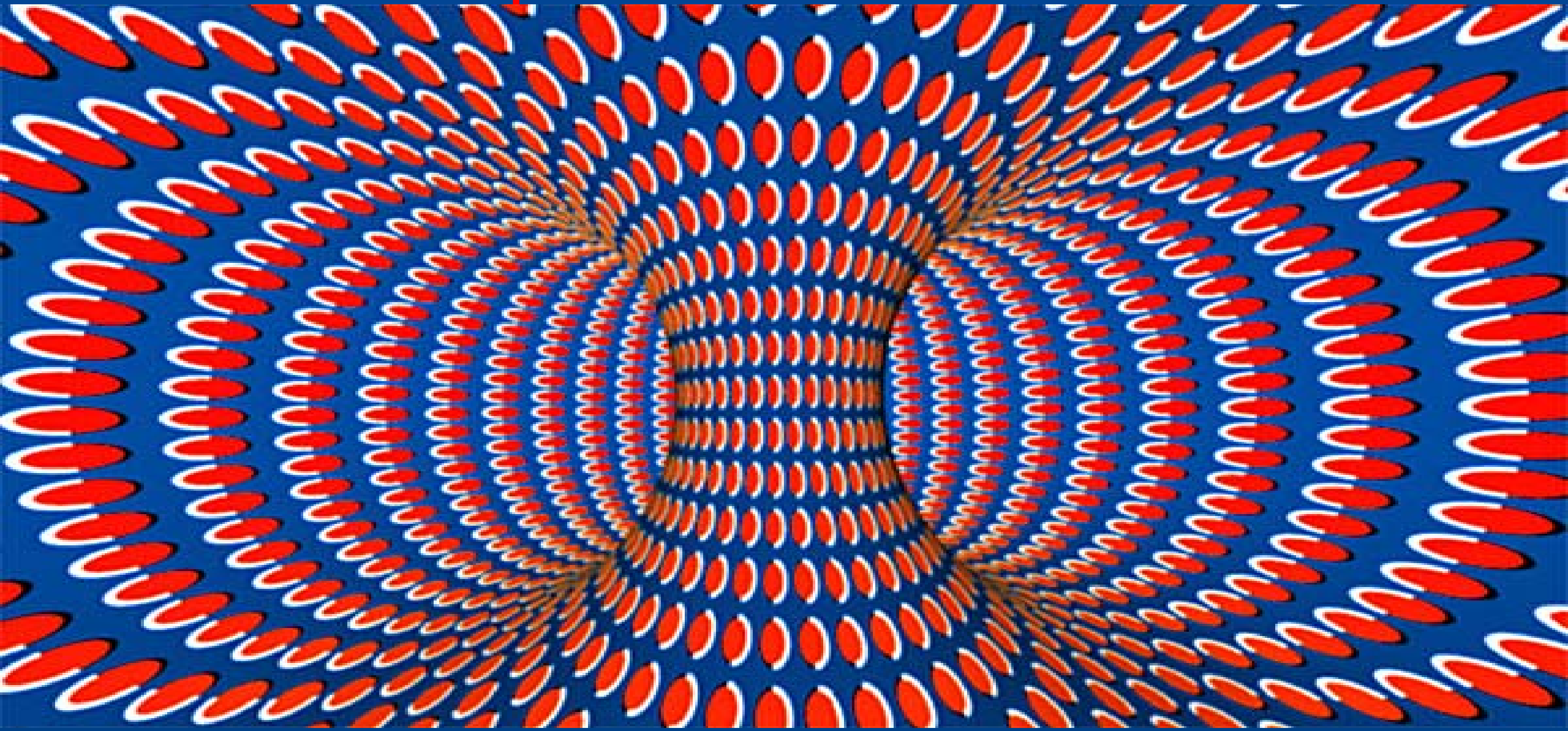
“Rollers” by Akiyoshi Kitaoka [Kitaoka 18]

Examples of motion illusions



“Rotating full moons” by Akiyoshi Kitaoka [Kitaoka 18]

Examples of motion illusions



“Spinning Vortex” by Beau Deeley [Deeley 12]

Caveats and recommendations

- Be aware of human perception limitations
  - avoid using gray scale as a method for representing more than a few (two to four) numerical values
    - close shades of gray are distinguishable only if they touch
  - beware of Mach bands artifacts
  - beware of motion illusions
- Exploit perception illusions
  - stress area borders (tricking DoG)
  - consider using Cornsweet contours instead of simple lines to define bounded regions

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Caveats and recommendations

- Simplify the representations
  - as human perception is limited, it makes no sense to fill an interface with overwhelming graphics
  - the “form”, i.e., not indispensable graphics, should be limited down to a minimum
  - only “functional” graphics should be allowed?

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White spaces versus graphics

- More graphics
  - conveys more information
  - clutters the visualization
  - discourages the users
- More white spaces
  - conveys less information
  - provides a terse, succint, incisive visualization
  - wastes space

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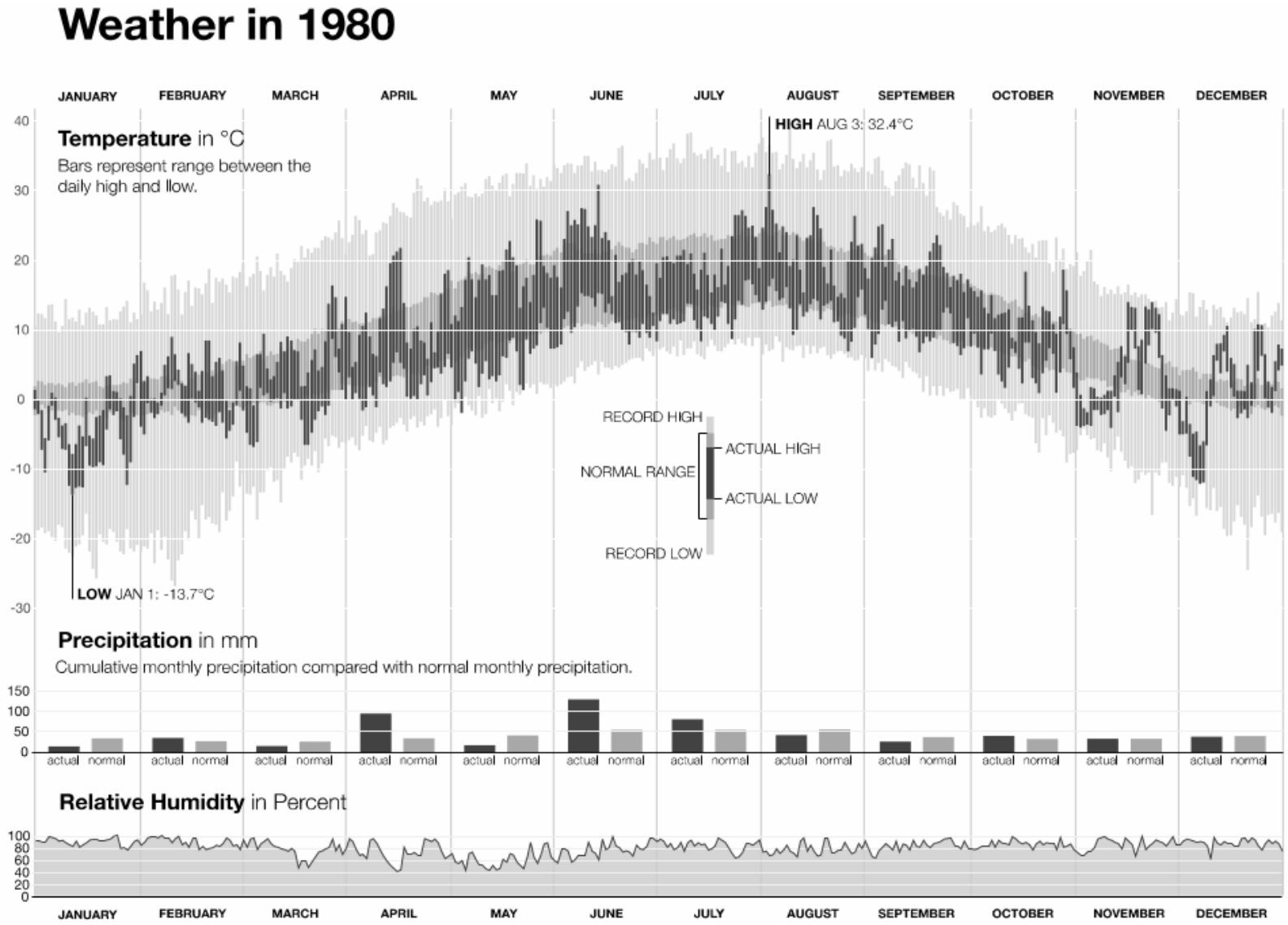
Maximize data density

- Principle from [Turfe 83]
- Data density
  - number of items/area of graphic
- This principle is controversial
  - white space may contribute to good visual design
- Analogous principle for displays
  - Pixels are precious! [Munzner 14]

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An example of high density

- Good data-ink ratio, uses small multiples, high data density
  - from [Turfe 83]
  - generated with the Protovis toolkit
  - printed annually by New York Times for more than 30 years



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Labeling and grid tips

- Avoid separate legends and keys
  - just have that information in the graphic
- Make grids, labeling, etc., very faint
  - so that they recede into background
- Let the information pop out

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Let the information pop out

- Before

Train No.	3701	XM 3301	3801	A 67	3 3803	3 3201	A3 51	3 3703	3 3807	3 3203	A3 61	3 3809	A3 47	3 3901	3 3811	3 3903	3 3813	3205	3815	3817	3819	3207	3821	3823	3825	3209	3827	3829	3831
New York, N.Y.	A.M. 12.10	A.M. 12.40	A.M. 1.30	A.M. 3.52	A.M. 4.50	A.M. 6.10	A.M. 6.25	A.M. 6.35	A.M. 6.50	A.M. 7.10	A.M. 7.30	A.M. 7.33	A.M. 7.45	A.M. 7.50	A.M. 8.05	A.M. 8.25	A.M. 8.40	A.M. 8.50	A.M. 9.10	A.M. 9.40	A.M. 10.10	A.M. 10.25	A.M. 10.40	A.M. 11.10	A.M. 11.40	A.M. 11.50	P.M. 12.10	P.M. 12.40	P.M. 1.10
Newark, N.J. P	12.24	12.55	1.44	4.07	5.04	6.24	6.38	6.49	7.04	7.24	7.45	7.47	7.59	8.04	8.19	8.39	8.54	9.04	9.24	9.54	10.24	10.39	10.54	11.24	11.54	12.04	12.24	12.54	1.24
North Elizabeth	....	....	....	....	....	....	....	....	....	7.30	....	....	....	8.10	....	....	....	....	....	....	....	....	....	....	....	....	....	....	....
Elizabeth	12.31	1.03	1.51	....	5.11	6.31	....	6.56	7.11	7.32	....	7.54	....	8.13	8.26	8.46	9.01	9.11	9.31	10.01	10.31	10.46	11.01	11.31	12.01	12.11	12.31	1.01	1.31
Linden	12.36	....	1.56	....	5.16	6.36	....	7.01	7.15	7.37	....	7.59	....	8.18	8.31	8.51	9.06	....	9.36	10.06	10.36	....	11.06	11.36	12.06	....	12.36	1.06	1.36
North Rahway	....	....	....	....	....	....	....	7.03	....	7.39	....	....	....	8.20	8.33	8.54	....	....	....	....	....	....	....	....	....	....	....	....	....
Rahway	12.40	1.11	2.00	....	5.20	6.40	....	7.06	7.20	7.42	....	8.03	....	8.24	8.36	8.57	9.10	9.18	9.40	10.10	10.40	10.53	11.10	11.40	12.10	12.18	12.40	1.10	1.40
Metro Park (Iselin)	12.44	....	2.04	4.26	5.24	....	6.56	7.10	7.25	....	8.04	8.07	8.15	....	8.40	....	9.14	....	9.44	10.14	10.44	....	11.14	11.44	12.14	....	12.44	1.14	1.44
Metuchen	12.48	....	2.08	....	5.28	....	....	7.14	7.29	....	....	8.11	....	....	8.44	....	9.18	....	9.48	10.18	10.48	....	11.18	11.48	12.18	....	12.48	1.18	1.48
Edison	12.51	....	2.11	....	....	....	....	7.17	7.32	....	....	8.14	....	....	8.47	....	9.21	....	....	10.21	....	....	11.21	....	12.21	....	....	1.21	....
New Brunswick	12.55	....	2.15	....	5.35	....	7.05	7.21	7.35	....	....	8.18	8.25	....	8.50	....	9.25	....	9.54	10.25	10.54	....	11.25	11.54	12.25	....	12.54	1.25	1.54
Jersey Avenue	1.02	....	2.18	....	....	....	....	7.28	....	....	....	8.21	....	....	....	....	9.28	....	....	10.28	....	....	11.28	....	12.28	....	....	1.28	....
Princeton Jct. S	....	....	2.31	....	5.50	....	7.19	....	7.50	....	....	8.34	8.41	....	9.05	....	9.41	....	10.09	10.41	11.09	....	11.41	12.09	12.41	....	1.09	1.41	2.09
Trenton, N.J.	....	....	2.42	4.58	6.03	....	7.28	....	8.01	....	8.31	8.44	8.52	....	9.16	....	9.52	....	10.19	10.52	11.19	....	11.52	12.19	12.52	....	1.22	1.52	2.20

- After

	am ●																										
New York, NY	12.10	12.40	1.30	3.52	4.50	6.10	6.25	6.35	6.50	7.10	7.30	7.33	7.45	7.50	8.05	8.25	8.40	8.50	9.10	9.40	10.10	10.25	10.40	11.10	11.40		
Newark, NJ[P]	12.24	12.55	1.44	4.07	5.04	6.24	6.38	6.49	7.04	7.24	7.45	7.47	7.59	8.04	8.19	8.39	8.54	9.04	9.24	9.54	10.24	10.39	10.54	11.24	11.54		
North Elizabeth										7.30	..	..	..	8.10	..	..	..	..	..	..	..	..	..	..	..		
Elizabeth	12.31	1.03	1.51	..	5.11	6.31	..	6.56	7.11	7.32	..	7.54	..	8.13	8.26	8.46	9.01	9.11	9.31	10.01	10.31	10.46	11.01	11.31	12.01		
Linden	12.36	..	1.56	..	5.16	6.36	..	7.01	7.15	7.37	..	7.59	..	8.18	8.31	8.51	9.06	..	9.36	10.06	10.36	..	11.06	11.36	12.06		
North Rahway								7.03	..	7.39	..	..	..	8.20	8.33	8.54	..	..	..	..	..	..	..	..	..		
Rahway	12.40	1.11	2.00	..	5.20	6.40	..	7.06	7.20	7.42	..	8.03	..	8.24	8.36	8.57	9.10	9.18	9.40	10.10	10.40	10.53	11.10	11.40	12.10		
Metro Park (Iselin)	12.44		2.04	4.26	5.24		6.56	7.10	7.25		8.04	8.07	8.15		8.40		9.14		9.44	10.14	10.44		11.14	11.44	12.14		
Metuchen	12.48		2.08	..	5.28		..	7.14	7.29		..	8.11	..		8.44		9.18		9.48	10.18	10.48		11.18	11.48	12.18		
Edison	12.51		2.11	..	..		..	7.17	7.32		..	8.14	..		8.47		9.21		..	10.21	..		11.21	..	12.21		
New Brunswick	12.55		2.15	..	5.35		7.05	7.21	7.35		..	8.18	8.25		8.50		9.25		9.54	10.25	10.54		11.25	11.54	12.25		
Jersey Avenue	1.02		2.18	..	..		..	7.28	..		..	8.21	..		..		9.28		..	10.28	..		11.28	..	12.28		
Princeton Junction[S]			2.31	..	5.50		7.19		7.50		..	8.34	8.41		9.05		9.41		10.09	10.41	11.09		11.41	12.09	12.41		
Trenton, NJ			2.42	4.58	6.03		7.28		8.01		8.31	8.44	8.52		9.16		9.52		10.19	10.52	11.19		11.52	12.19	12.52		
TRAIN NUMBER	3701	3301	3801	67	3803	3201	51	3703	3807	3203	61	3809	47	3901	3811	3903	3813	3205	3815	3817	3819	3207	3821	3823	3825		
NOTES		XM		⟝	3	3	⟝3	3	3	3	⟝3	3	⟝3	3	3	3	3										

- From [Turfe 90]

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Form versus function

- Form = graphics that do not encode data
  - embellishments of the drawings or interfaces
  - threedimensional effects
  - shadows and lightning
  - …
- Function = graphics that encode data
  - meaningful shapes
  - meaningful colors
  - meaningful sizes
    - lengths, areas, volumes
  - …
- A good design is always a mix of form and function

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Function priority

- “Function first, form next” [Munzner 14]
  - once you have fuction, you can refine your design to reach a good form also
  - if you get form first, it would be difficult to add function next

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Turfe's data-ink ratio principle

- Maximize the data-ink ratio [Turfe 83]

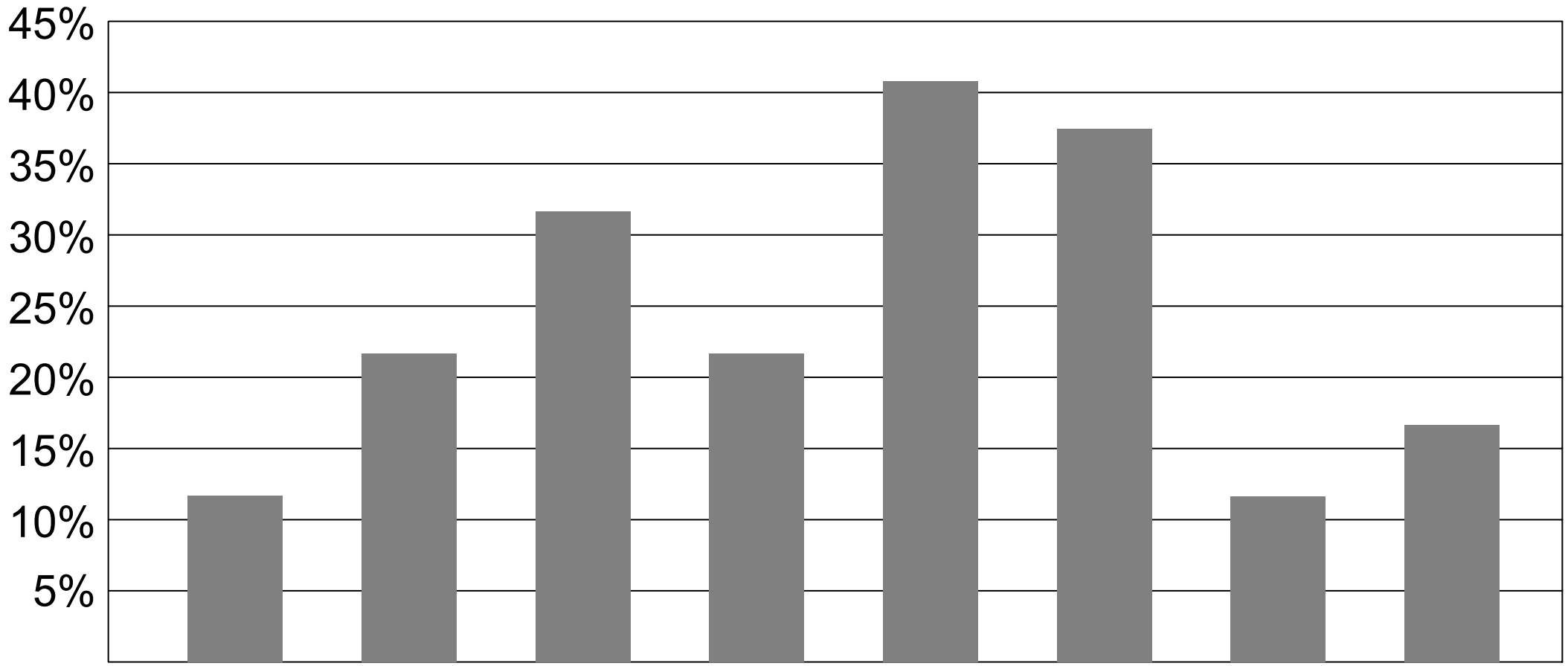
$$\text{data-ink ratio} = \frac{\text{data-ink}}{\text{total ink used in graphic}}$$

- Goals
  - limit the amount of graphics
  - produce plain, clear, "minimialist" graphs
  - avoid "chartjunk", i.e, extraneous visual elements that distract from message
    - "if the numbers are boring, then you've got the wrong numbers" [Turfe 90]

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Example of ink minimization

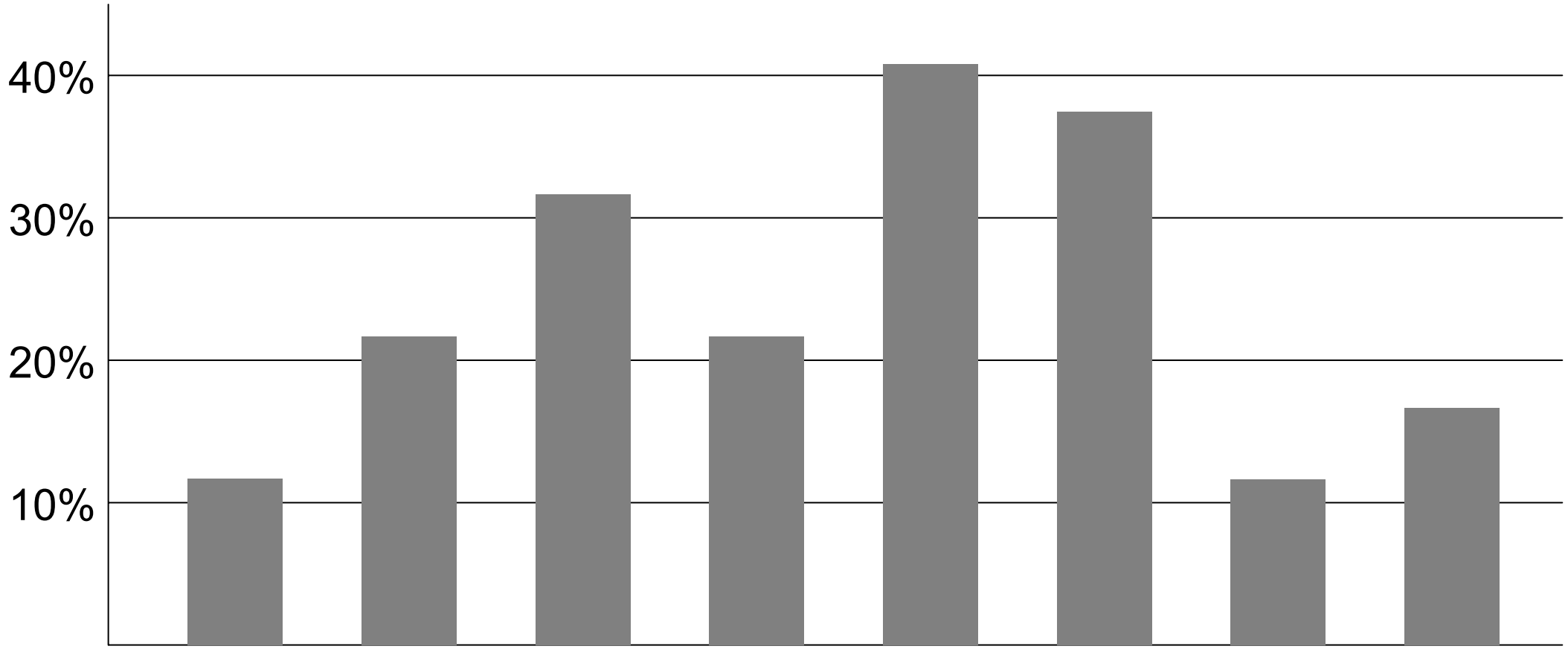
- From [Stasko 13]



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Example of ink minimization

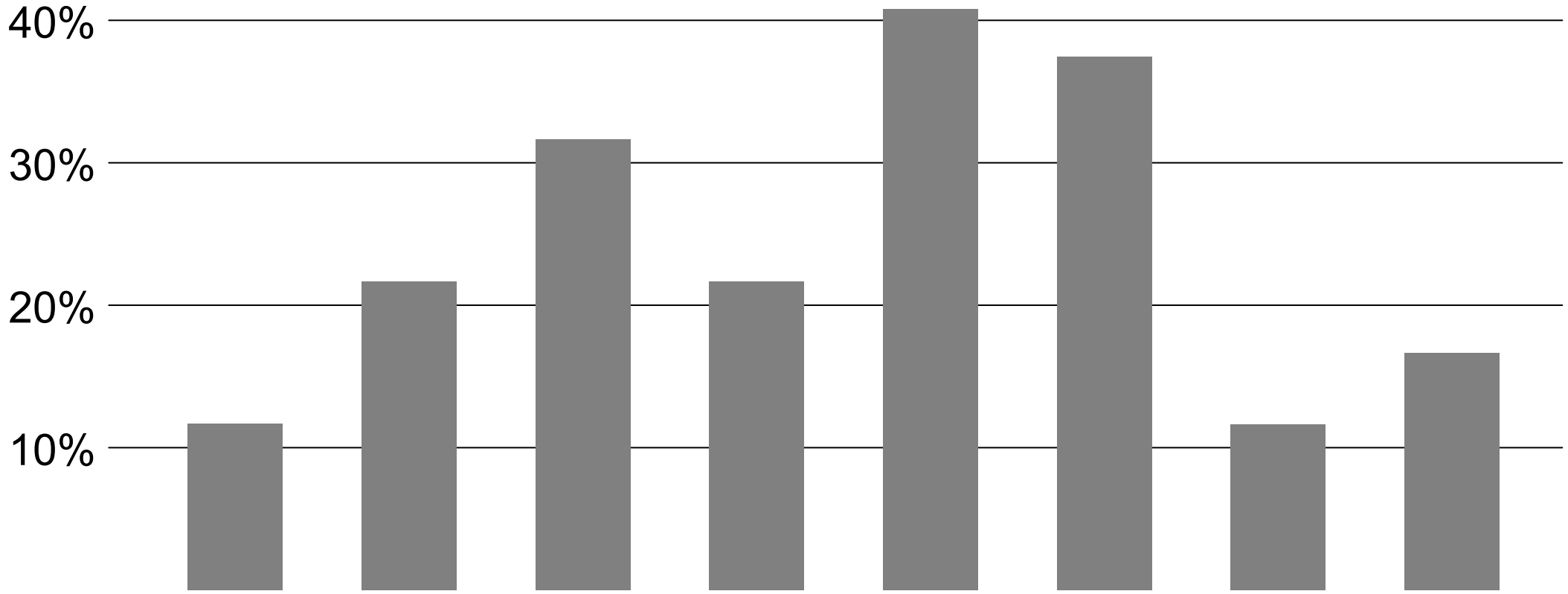
- From [Stasko 13]



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Example of ink minimization

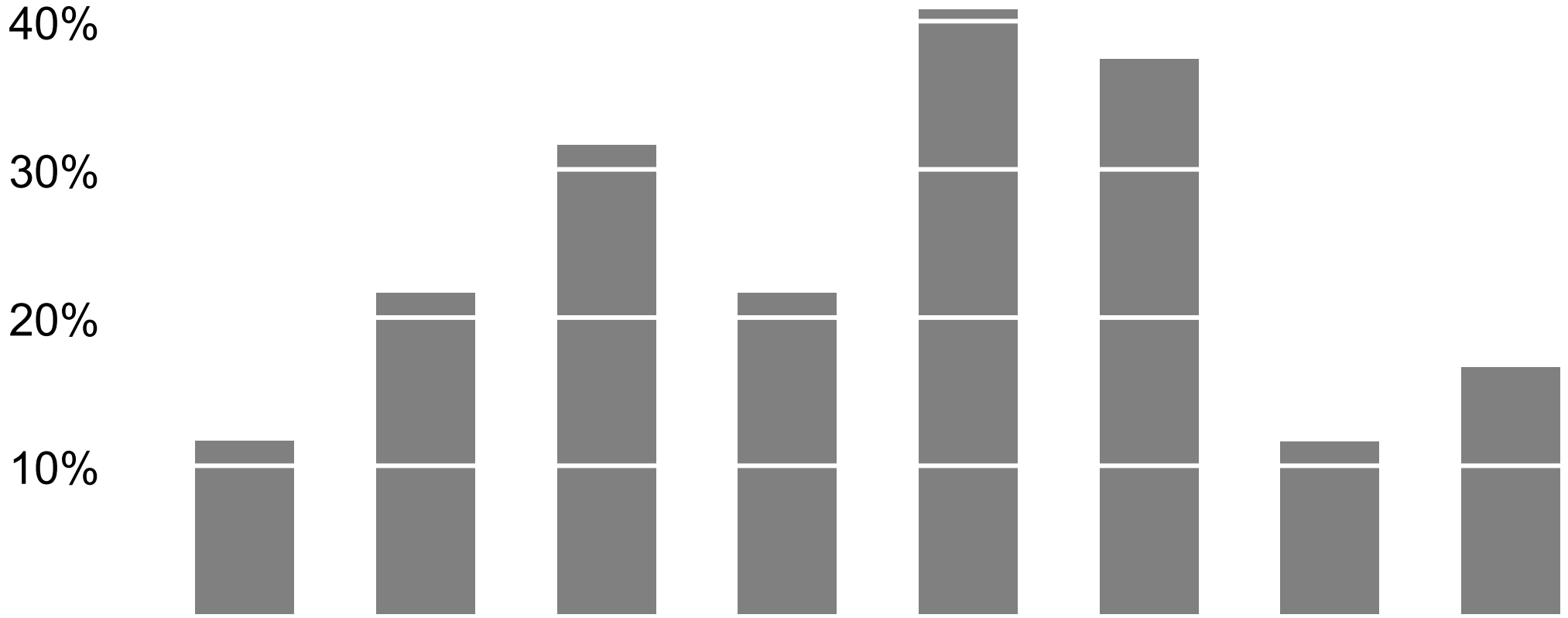
- From [Stasko 13]



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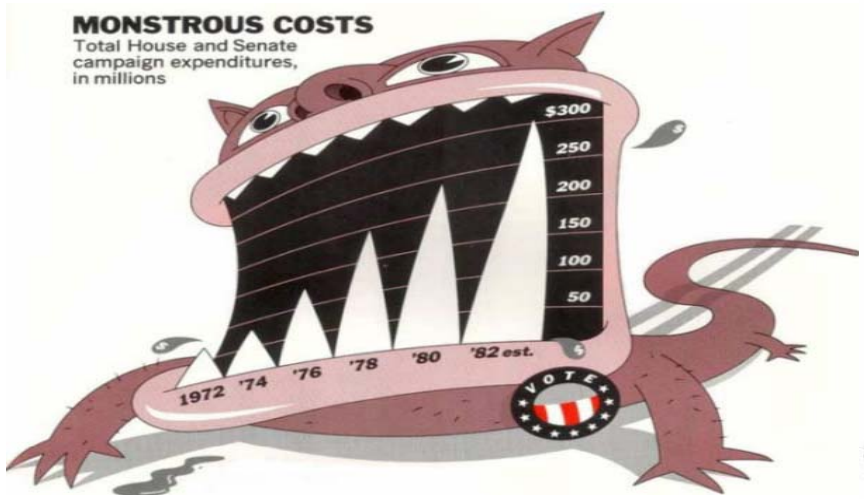
Example of ink minimization

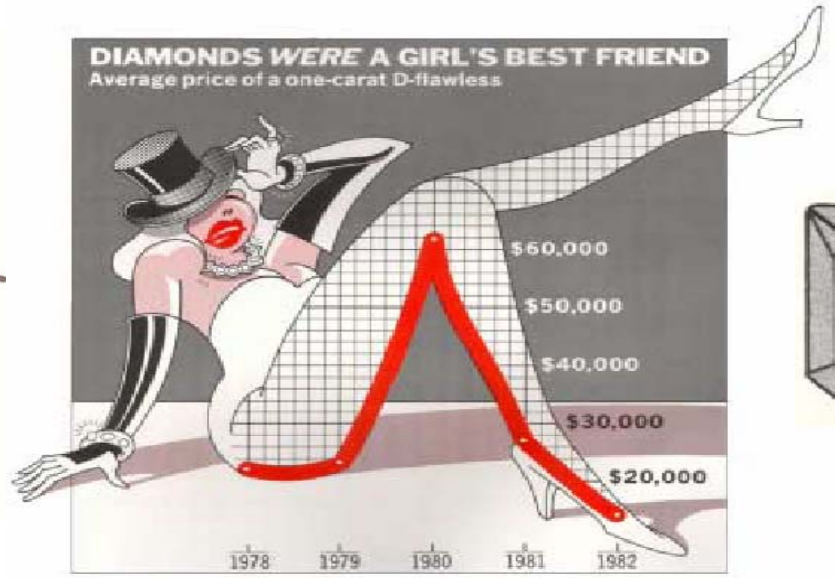
- From [Stasko 13]

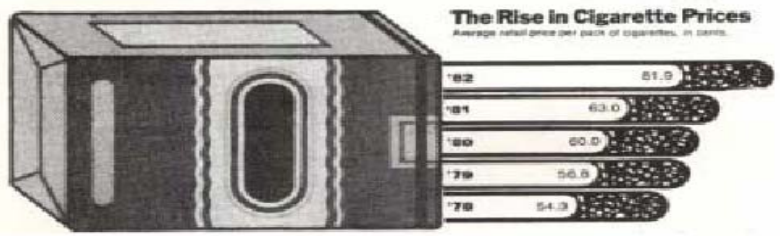


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Always avoid chartjunk?





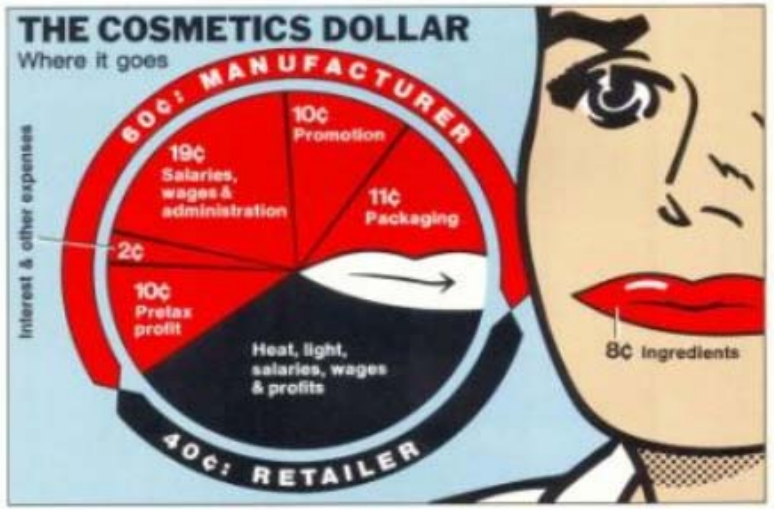


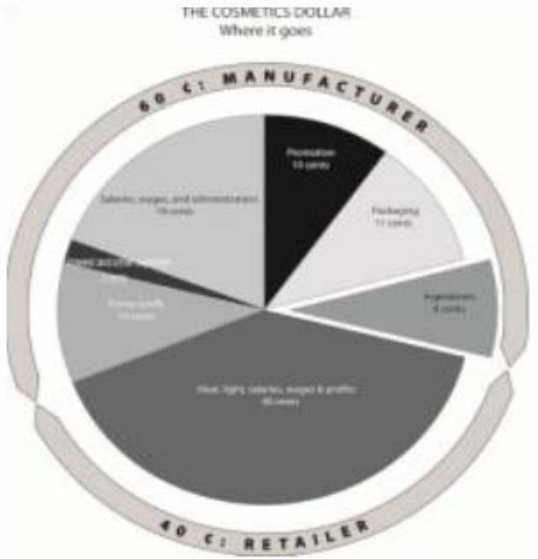
Pictures from [Bateman *et al*., '10]

- Turfe's ink-ratio principle rules out chartjunk on the basis that they reduce the effectiveness of graphics
  - recent studies, however, point in the opposite direction

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Sometimes chartjunk are effective



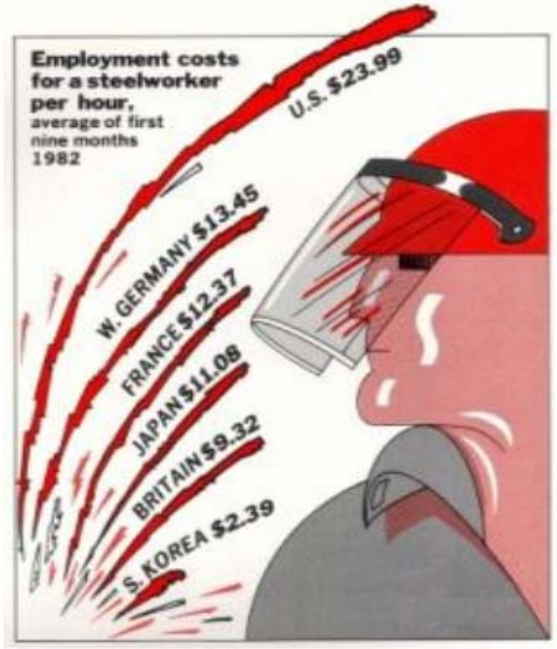


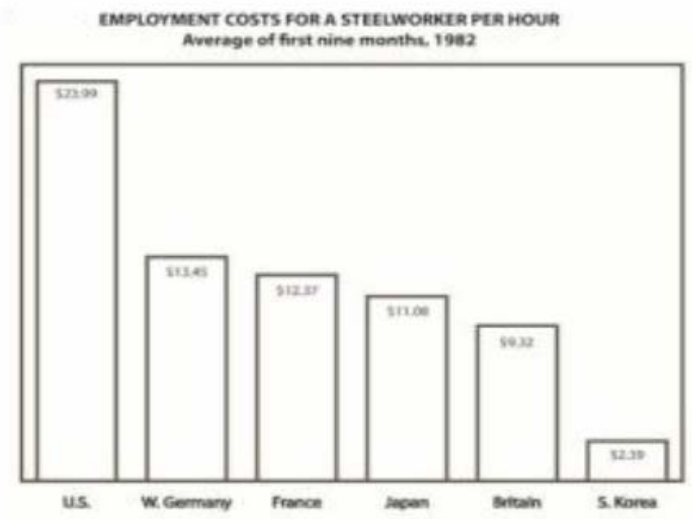
Pictures from [Bateman *et al*., '10]

- By comparing embellished charts with standard ones [Bateman *et al*., 10] show that
  - people's accuracy in describing the embellished charts is no worse than for plain charts
  - immediate recall is also unaffected
  - recall after a two-to-three-week gap is significantly better
  - embellished charts are preferred by common people

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Minimalist Vs embellished charts





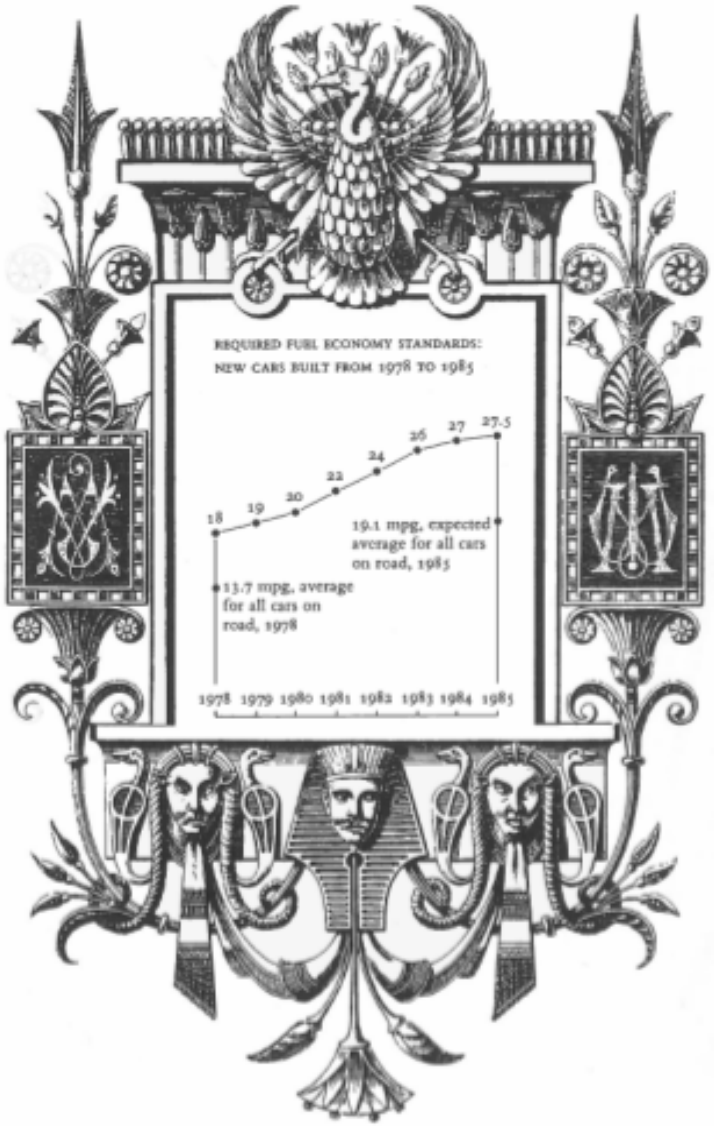
Pictures from [Bateman *et al*., '10]

- Plain charts tend to be uninteresting and unexciting
  - this is why their long term memory fades away
- Possible compromise
  - avoid chartjunk but don't be bland

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Turfe on decorations

- Turfe himself is not against editorial decorations provided they do not interfere with data representation
  - picture from [Turfe 83]



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Form Vs function: conclusions

- Function priority
  - function first, form next [Munzner 14]
  - a radical view:
    - Turfe's data-ink ratio minimization [Turfe 83]
    - multifunctioning graphical elements
- Form importance
  - form affects preference and memorability
- A good balance between form and function has to be pursued

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